Functional Requirements Document (FRD)

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Functional Requirements Document – Overall (all modules) · **Version:** 1.0 · **Status:** Draft **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026 **Author:** Jojo George (jojogeorge3344@gmail.com) **Based on:** BRD-TaskFlow-Overall.md v1.0 (and the 16 sub-module BRDs v1.0) © 2026 Jojo George. All rights reserved.

Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version of the overall FRD: system overview, module map, roles and the full permission matrix, cross-module workflows, system-level notification and message catalogue, global NFRs, full functional detail for Authentication and Legal, index of the 16 sub-module FRDs.

Related Documents

Document	Purpose	Location
BRD – Overall	Business needs, global rules `BR-nn`, global acceptance criteria `AC-nn`, `REQ-AUTH-nn`, `REQ-LEGAL-nn`	BRD-TaskFlow-Overall
BRDs – sub-modules (16)	`REQ-<CODE>-nn` baseline of every other screen	`../../BRD/<Folder>/BRD-<File stem>.md` (see section 2.3)
FRDs – sub-modules (16)	Detailed functional behaviour of every other screen	Section 2.3 (links relative to this file)
Development Plan	Screen-wise build plan (20 screens)	`TF - Documents/Screens Wise Feature Development Plan/`
Module workflow docs	Verified functional and technical behaviour, written from the code	`TaskFlow Proj/TFWorkflow/App WF in MD/`, `TaskFlow Proj/TFTechnicalWorkflow/Tech WF in MD/` (Authentication: `01-Auth`, Legal: `10-Legal`)
HLD / LLD / TDD	Technical design	`../../HLD/00-Overall/HLD-TaskFlow-Overall.md`, `../../LLD/00-Overall/LLD-TaskFlow-Overall.md`, `TF - Documents/TF - TDD/TDD-TaskFlow.md`

How this document is organised

This overall FRD holds what is shared by the whole system: modules, roles and permissions, cross-module workflows, the system-level notification and message catalogue, global non-functional requirements and the

complete functional detail of **Authentication** (`FR-AUTH-nn` , `UC-AUTH-nn`) and **Legal** (`FR-LEGAL-nn` , `UC-LEGAL-nn`), which have no sub-module FRD. Every other screen is specified in its sub-module FRD (`FR-<CODE>-nn` , `UC-<CODE>-nn`), indexed in section 2.3; that detail is not repeated here.

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1. Introduction

1.1 Purpose

This FRD specifies the testable functional behaviour of TaskFlow at system level: which modules exist, who may use what, how modules work together, which notifications and messages the system produces, and – in full – how sign-in, two-step verification, account recovery, security setup, sessions and the legal pages behave. It turns the overall BRD into "The system shall …" statements that developers implement and testers verify.

1.2 Scope

In scope of this document	Covered in sub-module FRDs
System overview, module map, sub-module FRD index	Dashboard (DASH)
Roles and the 64-entry permission matrix	Task Management incl. comments and activity (TASK), Employee Workload (WORK), Board (BOARD)
Cross-module workflows (task lifecycle, reminders, sign-in, recovery, MFA reset, role change, TFA actions, deactivation)	Employee (EMP), Project incl. positions (PROJ), Department (DEPT) Management
System-level notification, mail and message catalogue	Mail (MAIL), Notification incl. reminder engine (NOTIF)
Global NFRs (functional impact)	Reports (RPT)
Authentication: Login, Two-step verification step, Account Recovery, Security Setup, Sessions and Sign out, access control	Reminder (REM), Application (APP), TFA (TFAS) Settings, Role Configuration (ROLE)
Legal: five legal document pages, legal footer and ownership notice	My Profile incl. change password and two-step verification (PROF); TFA Assistant (TFA)

Out of scope (BRD section 3.2): external e-mail or SMS (SMTP of original Screen 14), multiple organisations, more than one application server, automatic scheduled backups, editing legal documents in the application, password reset by e-mail link or by an administrator choosing a password.

1.3 Audience

Developer (implementation), tester (test design from FR, UC and messages), reviewer, product owner (Jojo George).

1.4 Conventions

- Requirement IDs: `FR-<CODE>-nn` (this document: `FR-AUTH-nn` , `FR-LEGAL-nn`). Use cases: `UC-<CODE>-nn` . Global business rules `BR-nn` , global NFRs `NFR-nn` and acceptance criteria `AC-nn` come from the overall BRD; BRD requirements are `REQ-<CODE>-nn` .
- Priority: **Must / Should / Could** (MoSCoW), taken from the traced BRD requirement.
- "The system shall …" = mandatory behaviour; "should" = recommended.
- Permission matrix values: **Always On** (every role, cannot be removed), **Granted** (default on), **Denied** (default off), **Configurable** (custom roles: as granted by an administrator), **Not assignable** (Administrator only).
- "Public" = no sign-in and no permission key needed.
- Real UI labels are in **bold**; real messages in "quotes".

1.5 Definitions & Abbreviations

Term	Meaning
Scope	Data visibility: Own (own records), Team (a Manager's team), Organisation (everything).
Team	For a Manager: active people in a department the manager manages or on a project where the manager is project manager.
MFA / two-step verification	Sign-in with password plus a 6-digit time-based code from an authenticator app.
Backup code	One of 10 single-use codes in the form `XXXXX-XXXXX`.
Sign-in step (challenge)	Short-lived (5 minutes) proof that the password was correct, needed for the code step.
Access pass	Short session credential (30 minutes) sent with every request.
Renewal cookie	7-day, HttpOnly, Secure, SameSite Strict cookie used to renew the access pass; replaced at every renewal.
Security stamp	Account value that changes after a password change, password reset or MFA reset; old sessions stop working.
Reset permission	Single-use permission, valid 10 minutes, to set a new password after account recovery.
Setup pending	The user must still set up two-step verification (required by default).
Owner record	The single set of ownership values (owner name, year, contact email, rights statement, application name) used by the legal pages and the footer.
TFA	TaskFlow Assistant.
In QA	Task status for finished work that is being checked.

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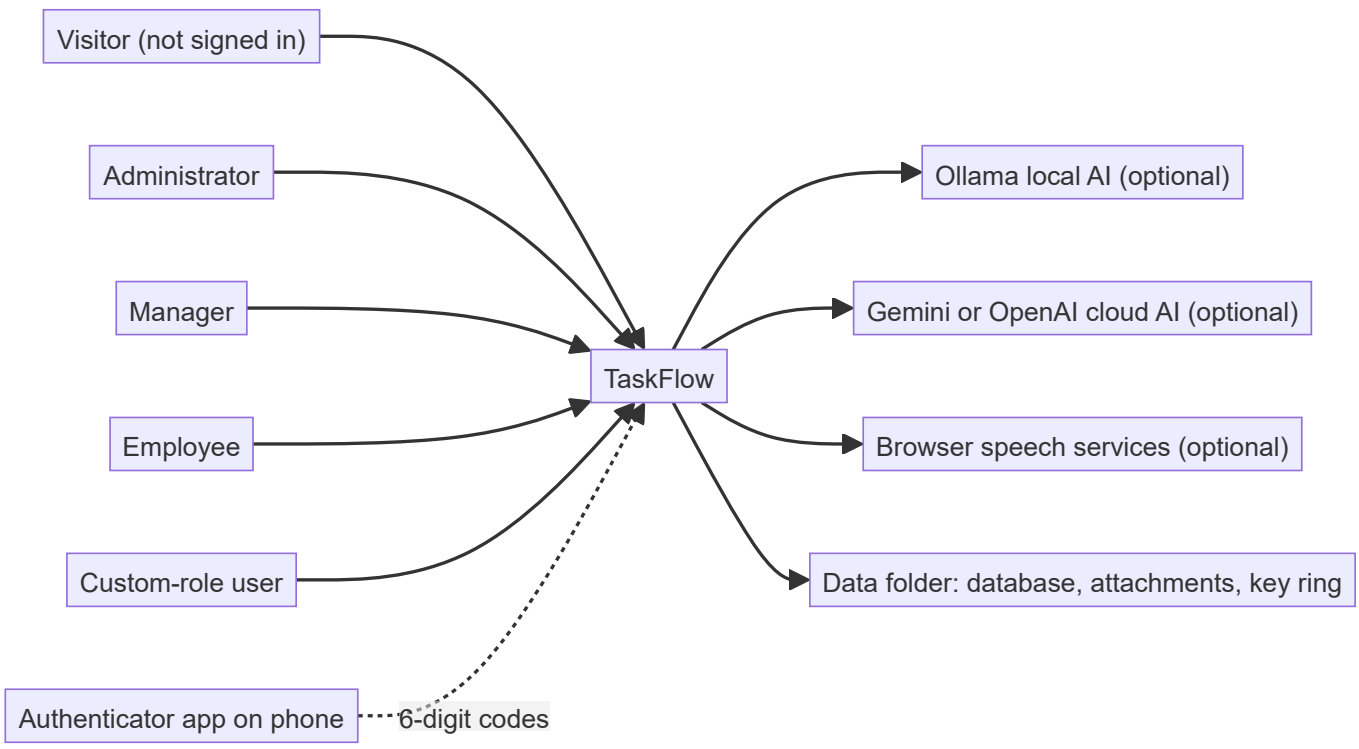
2. System Overview

2.1 Module Map

Code	Module	Child modules / screens	Primary roles	Specified in
AUTH	Authentication	Login, Two-step verification step, Account Recovery, Security Setup, Sessions and Sign out	Everyone (public), all signed-in users	This document, 4.1
DASH	Dashboard	Administrator, Manager and Employee Dashboard, analytics, filters, drill-down	All signed-in users	FRD-Dashboard
TASK	TaskFlow Monitoring › Task Management	Task List, Create Task, Task Details, Reassign Task, Task Discussion, Activity History	All signed-in users (scope by permission)	FRD-Task-Management
WORK	TaskFlow Monitoring › Employee Workload	Employee Workload	Administrator, Manager	FRD-Employee-Workload
BOARD	TaskFlow Monitoring › Board	Kanban Board	All signed-in users	FRD-Board
EMP	TaskFlow Management › Employee Management	Employee list, details panel (incl. **Reset MFA**), add/edit	Administrator (Manager view only)	FRD-Employee-Management
PROJ	TaskFlow Management › Project Management	Project Management, Project Details, team and positions	Administrator (Manager view only)	FRD-Project-Management

Code	Module	Child modules / screens	Primary roles	Specified in
DEPT	TaskFlow Management › Department Management	Department list, add/edit, members	Administrator (Manager view only)	FRD-Department-Management
MAIL	Communication › Mail	Mail folders, reading pane, Compose panel	All signed-in users	FRD-Mail
NOTIF	Communication › Notification	Notification Center, header bell, reminder engine	All signed-in users; System	FRD-Notification
RPT	Reports	Task Reports, CSV export	Administrator, Manager	FRD-Reports
REM	Settings › Reminder Settings	Reminder Settings	Administrator, Manager	FRD-Reminder-Settings
APP	Settings › Application Settings	Application Settings	Administrator	FRD-Application-Settings
TFAS	Settings › TFA Settings	TFA Settings (AI providers, models, keys, connection test)	Administrator	FRD-TFA-Settings
ROLE	Settings › Role Configuration	Role Configuration	Administrator	FRD-Role-Configuration
PROF	My Profile	My Profile, Change Password, Two-step verification (self-service)	All signed-in users	FRD-My-Profile
TFA	TFA Assistant	TFA Panel, How-to Help, Action Card, Voice controls	All signed-in users	FRD-Assistant
LEGAL	Legal	Copyright, Terms of Use, Privacy Policy, Disclaimer, Licences pages; legal footer	Everyone (public)	This document, 4.2

2.2 Context Diagram



TaskFlow sends nothing by external e-mail or SMS (BR-23). The authenticator app works offline on the user's phone; the only outbound connections are to a cloud AI provider when an administrator chooses one.

2.3 Sub-module FRD Index

#	Code	Module	Sub-module	Sub-module FRD	BRD baseline
0	ALL	(all)	Overall, incl. Authentication and Legal	This document	BRD-TaskFlow-Overall – REQ-AUTH-01 – 41, REQ-LEGAL-01 – 15
1	DASH	Dashboard	Dashboard	FRD-Dashboard	REQ-DASH-01 – 58
2	TASK	TaskFlow Monitoring	Task Management (incl. comments and activity history)	FRD-Task-Management	REQ-TASK-01 – 58
3	WORK	TaskFlow Monitoring	Employee Workload	FRD-Employee-Workload	REQ-WORK-01 – 17
4	BOARD	TaskFlow Monitoring	Board	FRD-Board	REQ-BOARD-01 – 18
5	EMP	TaskFlow Management	Employee Management	FRD-Employee-Management	REQ-EMP-01 – 49
6	PROJ	TaskFlow Management	Project Management (incl. positions)	FRD-Project-Management	REQ-PROJ-01 – 37
7	DEPT	TaskFlow Management	Department Management	FRD-Department-Management	REQ-DEPT-01 – 27
8	MAIL	Communication	Mail	FRD-Mail	REQ-MAIL-01 – 47
9	NOTIF	Communication	Notification (incl. reminder engine)	FRD-Notification	REQ-NOTIF-01 – 45
10	RPT	Reports	Reports	FRD-Reports	REQ-RPT-01 – 39
11	REM	Settings	Reminder Settings	FRD-Reminder-Settings	REQ-REM-01 – 22
12	APP	Settings	Application Settings	FRD-Application-Settings	REQ-APP-01 – 22
13	TFAS	Settings	TFA Settings	FRD-TFA-Settings	REQ-TFAS-01 – 25
14	ROLE	Settings	Role Configuration	FRD-Role-Configuration	REQ-ROLE-01 – 25
15	PROF	My Profile	My Profile (profile, change password, two-step verification)	FRD-My-Profile	REQ-PROF-01 – 29
16	TFA	TFA Assistant	Assistant	FRD-Assistant	REQ-TFA-01 – 45

Each sub-module FRD numbers its requirements `FR-<CODE>-nn`, traces each one to its BRD `REQ-<CODE>-nn`, and applies the global rules `BR-nn` of the overall BRD.

2.4 Screen and Route Map

Screen	Route / location	Module	Access
Login (with Two-step verification step)	`/login` (also the target of `/` and of any unknown address)	AUTH	Public
Forgot your password?	`/account-recovery`	AUTH	Public
Let's secure your account	`/security-setup`	AUTH	Signed in, setup pending
Sign out	Header → account area → **Sign out** icon	AUTH	Signed in
Administrator / Manager / Employee Dashboard	`/admin`, `/manager`, `/me`	DASH	`dashboard.view.organization`, `dashboard.view.team`, `dashboard.view.own`
Task List, Create Task, Task Details	`/tasks`, `/tasks/new`, `/tasks/:id`	TASK	`tasks.view.own` / `tasks.view.all`, `tasks.create`
Task Discussion, Activity History, Reassign Task	`/tasks/:id/discussion`, `/tasks/:id/activity`, `/tasks/:id/reassign`	TASK	Task access; `tasks.activity.view`; `tasks.reassign`
Board	`/board`	BOARD	`board.view`
Employee Workload	`/workload`	WORK	`workload.view`
Employee, Project, Department Management	`/employees`, `/projects`, `/projects/:id`, `/departments`	EMP, PROJ, DEPT	Matching view and action permissions
Mail, Notification Center	`/mail`, `/notifications`	MAIL, NOTIF	`mail.read`, `notifications.view`
Task Reports	`/reports`	RPT	`reports.view`
Reminder, Application, TFA Settings, Role Configuration	`/settings/reminders`, `/settings/application`, `/settings/tfa`, `/settings/permissions`	REM, APP, TFAS, ROLE	Matching settings permissions
My Profile, Change Password	`/profile`, `/profile/change-password`	PROF	Always On
TFA Panel	TFA button on every signed-in screen	TFA	`assistant.chat`
Legal document pages	`/legal/copyright`, `/legal/terms`, `/legal/privacy`, `/legal/disclaimer`, `/legal/licenses`	LEGAL	Public
Legal footer	Bottom of Login, Account Recovery and every signed-in screen	LEGAL	Public

2.5 Change from Original BRD (system level)

Original BRD	Current functional requirement	Where
Screen 1 – password-only login	Password plus mandatory two-step verification; generic failure message; renewable sessions re-checked on every request	4.1.1 – 4.1.5
No account recovery	Self-service recovery with an authenticator code; nothing sent by e-mail or SMS	4.1.3

Original BRD	Current functional requirement	Where
No security setup	Mandatory Security Setup before any other screen	4.1.4
Fixed roles	Permission-based access from a fixed catalogue of 64 permissions; read-only system roles and custom roles	3.2, FRD-Role-Configuration
Screen 14 – SMTP e-mail settings	**Superseded** by internal Mail; reminders optionally copied into the Mail inbox	7, FRD-Mail, FRD-Reminder-Settings
Five task statuses	**In QA** added (six statuses, five Board columns)	5.1, FRD-Task-Management, FRD-Board
No legal pages	Five public legal documents and an ownership footer	4.2

3. User Roles & Permissions

3.1 Roles

Role	Description	Data scope
Visitor	Anyone not signed in: Login, Account Recovery, legal pages and footer only.	Public pages only
Administrator	System role, holds every permission; the role is re-granted every permission at each application start.	Organisation
Manager	System role for people who lead work: tasks (no delete), workload, reports, reminder settings, read-only master data.	Team for tasks and the team dashboard; Organisation for reports and workload
Employee	System role for people who carry out work: own tasks, status, hours, comments, board, Mail, notifications, profile, TFA.	Own
Custom Role	Created by an administrator from the fixed catalogue; Always On permissions included; "Give the Administrator role" never assignable.	As granted: Own, Team or Organisation

System roles cannot be changed (BR-05). Every user holds exactly one role. Role and permission changes apply on the user's next request and are picked up by the session check within 15 seconds (FR-AUTH-44, FR-AUTH-45).

3.2 Permission Matrix (functional view)

Source: permission catalogue, 64 entries. "Requires" = prerequisite permission that is ticked automatically and must be held as well.

Module › Sub-module › Function	Permission key	Requires	Administrator	Manager	Employee	Custom role
Authentication › Sign in, code step, recovery, renewal, sign out	(none – public)	–	Always On	Always On	Always On	Always On
Legal › Legal pages and footer	(none – public)	–	Always On	Always On	Always On	Always On
Dashboard › View my dashboard	`dashboard.view.own`	–	Always On	Always On	Always On	Always On
Dashboard › View team dashboard	`dashboard.view.team`	–	Granted	Granted	Denied	Configurable
Dashboard › View organization dashboard	`dashboard.view.organization`	–	Granted	Denied	Denied	Configurable

Module › Sub-module › Function	Permission key	Requires	Administrator	Manager	Employee	Custom role
Dashboard › View charts and reports	`dashboard.analytics.view`	–	Granted	Granted	Granted	Configurable
Dashboard › Use dashboard filters	`dashboard.filters.use`	–	Granted	Granted	Granted	Configurable
Task Management › View my tasks	`tasks.view.own`	–	Always On	Always On	Always On	Always On
Task Management › View all tasks	`tasks.view.all`	–	Granted	Granted	Denied	Configurable
Task Management › Create tasks	`tasks.create`	–	Granted	Granted	Denied	Configurable
Task Management › Delete tasks	`tasks.delete`	–	Granted	Denied	Denied	Configurable
Task Management › Edit tasks	`tasks.update`	–	Granted	Granted	Denied	Configurable
Task Management › Reassign tasks	`tasks.reassign`	–	Granted	Granted	Denied	Configurable
Task Management › Manage task recipients	`tasks.recipients.manage`	–	Granted	Granted	Denied	Configurable
Task Management › Post task comments	`tasks.comment.create`	–	Granted	Granted	Granted	Configurable
Task Management › Edit own task comments	`tasks.comment.edit`	–	Granted	Granted	Granted	Configurable
Task Management › Delete own task comments	`tasks.comment.delete`	–	Granted	Granted	Granted	Configurable
Task Management › View task history	`tasks.activity.view`	–	Granted	Granted	Granted	Configurable
Employee Workload › View workload	`workload.view`	–	Granted	Granted	Denied	Configurable
Employee Workload › Use workload filters	`workload.filters.use`	`workload.view`	Granted	Granted	Denied	Configurable
Employee Workload › Open an employee's tasks	`workload.tasks.open`	`workload.view`	Granted	Granted	Denied	Configurable
Board › View board	`board.view`	–	Granted	Granted	Granted	Configurable
Board › Use board filters	`board.filters.use`	`board.view`	Granted	Granted	Granted	Configurable
Board / Task Details › Update task status	`tasks.status.change`	–	Granted	Granted	Granted	Configurable
Employee Management › View employees	`employees.view`	–	Granted	Granted	Denied	Configurable
Employee Management › Create employees	`employees.create`	`employees.view`	Granted	Denied	Denied	Configurable

Module › Sub-module › Function	Permission key	Requires	Administrator	Manager	Employee	Custom role
Employee Management › Edit employees	`employees.edit`	`employees.view`	Granted	Denied	Denied	Configurable
Employee Management › Activate or deactivate employees	`employees.status.change`	`employees.view`	Granted	Denied	Denied	Configurable
Employee Management › Assign roles	`employees.role.assign`	`employees.view`	Granted	Denied	Denied	Configurable
Employee Management › Reset two-step verification	`employees.mfa.reset`	`employees.view`	Granted	Denied	Denied	Configurable
Employee Management › Delete employees	`employees.delete`	`employees.view`	Granted	Denied	Denied	Configurable
Project Management › View projects	`projects.view`	–	Granted	Granted	Denied	Configurable
Project Management › Create projects	`projects.create`	`projects.view`	Granted	Denied	Denied	Configurable
Project Management › Edit projects	`projects.edit`	`projects.view`	Granted	Denied	Denied	Configurable
Project Management › Delete projects	`projects.delete`	`projects.view`	Granted	Denied	Denied	Configurable
Project Management › Manage project members	`projects.members.manage`	`projects.view`	Granted	Denied	Denied	Configurable
Project Management › Create positions	`positions.create`	`projects.view`	Granted	Denied	Denied	Configurable
Department Management › View departments	`departments.view`	–	Granted	Granted	Denied	Configurable
Department Management › Create departments	`departments.create`	`departments.view`	Granted	Denied	Denied	Configurable
Department Management › Edit departments	`departments.edit`	`departments.view`	Granted	Denied	Denied	Configurable
Department Management › Delete departments	`departments.delete`	`departments.view`	Granted	Denied	Denied	Configurable
Department Management › Manage department members	`departments.members.manage`	`departments.view`	Granted	Denied	Denied	Configurable
Mail › Read mail	`mail.read`	–	Granted	Granted	Granted	Configurable
Mail › Write and send mail	`mail.compose`	–	Granted	Granted	Granted	Configurable
Mail › Organize mail	`mail.organize`	–	Granted	Granted	Granted	Configurable
Mail › Use mail attachments	`mail.attachments`	–	Granted	Granted	Granted	Configurable

Module › Sub-module › Function	Permission key	Requires	Administrator	Manager	Employee	Custom role
Notification › View notifications	`notifications.view`	–	Granted	Granted	Granted	Configurable
Notification › Mark notifications read	`notifications.manage`	–	Granted	Granted	Granted	Configurable
Reports › View reports	`reports.view`	–	Granted	Granted	Denied	Configurable
Reports › Export reports	`reports.export`	`reports.view`	Granted	Granted	Denied	Configurable
Settings › Reminder settings	`settings.reminders.manage`	–	Granted	Granted	Denied	Configurable
Settings › Application settings	`settings.application.manage`	–	Granted	Denied	Denied	Configurable
Settings › View AI provider settings	`settings.ai.view`	–	Granted	Denied	Denied	Configurable
Settings › Edit AI provider settings	`settings.ai.edit`	`settings.ai.view`	Granted	Denied	Denied	Configurable
Settings › Test AI providers	`settings.ai.test`	`settings.ai.view`	Granted	Denied	Denied	Configurable
Settings › Remove AI models	`settings.ai.model.remove`	`settings.ai.view`	Granted	Denied	Denied	Configurable
Role Configuration › Manage roles and permissions	`roles.manage`	–	Granted	Denied	Denied	Configurable
Role Configuration › View all roles	`roles.view.all`	–	Granted	Denied	Denied	Configurable
Role Configuration › Modify access	`roles.modify.access`	–	Granted	Denied	Denied	Configurable
Role Configuration › Give the Administrator role	`roles.administrator.assign`	–	Granted	Denied	Denied	Not assignable
My Profile › View my profile	`profile.view`	–	Always On	Always On	Always On	Always On
My Profile › Edit my personal information	`profile.edit`	–	Always On	Always On	Always On	Always On
My Profile › Change my password	`profile.password.change`	–	Always On	Always On	Always On	Always On
My Profile › Manage my authenticator (Security Setup, My Profile)	`profile.mfa.manage`	–	Always On	Always On	Always On	Always On
TFA Assistant › Chat with the assistant	`assistant.chat`	–	Granted	Granted	Granted	Configurable
TFA Assistant › Confirm assistant actions	`assistant.actions.confirm`	–	Granted	Granted	Granted	Configurable

Legacy keys. Older combined keys are accepted and expand to their fine-grained keys: `dashboard.admin` → `dashboard.view.organization`; `dashboard.manager` → `dashboard.view.team`; `projects.manage` → `projects.create`, `projects.edit`, `projects.delete`, `projects.members.manage`, `positions.create`; `departments.manage` → `departments.create`, `departments.edit`, `departments.delete`, `departments.members.manage`; `employees.manage` → `employees.create`, `employees.edit`, `employees.status.change`, `employees.mfa.reset`; `settings.ai.manage` → `settings.ai.view`, `settings.ai.edit`, `settings.ai.test`, `settings.ai.model.remove`. They are not shown as separate rows in Role Configuration.

3.3 Global Access-Control Behaviour

Layer	Behaviour
Menu	Shows only the entries the user's permissions allow (BR-04, NFR-12).
Screen address	Opening a screen without its permission sends the user to their own dashboard; no message is shown.
Server	Checks the permission on every request (BR-04); refused requests get "403 Forbidden" and the screen shows its fallback error. Another user's notification, mail or assistant conversation is reported as not found (BR-20).
Setup pending	Only `/security-setup` opens; the server refuses every function except sign-in, the own profile and the MFA setup functions (BR-03).

The detailed functional requirements for these layers are FR-AUTH-29, FR-AUTH-45, FR-AUTH-49 – FR-AUTH-51.

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4. Functional Requirements by Module

Full detail is given for **Authentication (4.1)** and **Legal (4.2)** only. Section 4.3 summarises the other modules and points to their FRDs.

4.1 Module: Authentication (AUTH)

Purpose: Let registered, active users sign in securely with password and authenticator code, recover a forgotten password without an administrator, secure their account before first use, and keep the session safe until sign-out. · **BRD reference:** Overall BRD SCREEN 1 – 3, section 4.3 – 4.4, REQ-AUTH-01 – REQ-AUTH-41, BR-01 – BR-04, BR-23 – BR-25, BR-28

4.1.1 Screen: Login (Sign in)

Field	Value
Route / location	`/login`; also opened for `/` and for any unknown address
Access	Public (Visitor). Completed by Administrator, Manager, Employee and Custom Role users with an active, not deleted account
Entry points	Opening TaskFlow; redirect from any protected screen when not signed in (with a return address); end of a session; ← **Back to sign in** on Account Recovery; **Back to sign in** on a legal page

Screen elements

Element	Type	Description	Mandatory	Visible to
"TF" logo, **TaskFlow**, "Sign in to continue"	Header text	Brand and subtitle	–	Everyone

Element	Type	Description	Mandatory	Visible to
Notice area	Info box	Reason the user was sent back to sign in (session ended, credentials changed, password reset)	–	Everyone, when a reason exists
Email	Text input	Email address or username; pre-filled with the identifier last used on this device	Yes	Everyone
Password	Password input	Password, with eye button **Show password / Hide password**	Yes	Everyone
Error area	Message	Field and sign-in errors	–	Everyone
Sign in	Button	Submits; shows "Signing in…" while working	–	Everyone
Forgot password?	Link	Opens Account Recovery	–	Everyone
Legal footer	Footer	Ownership line and five links (4.2.2)	–	Everyone

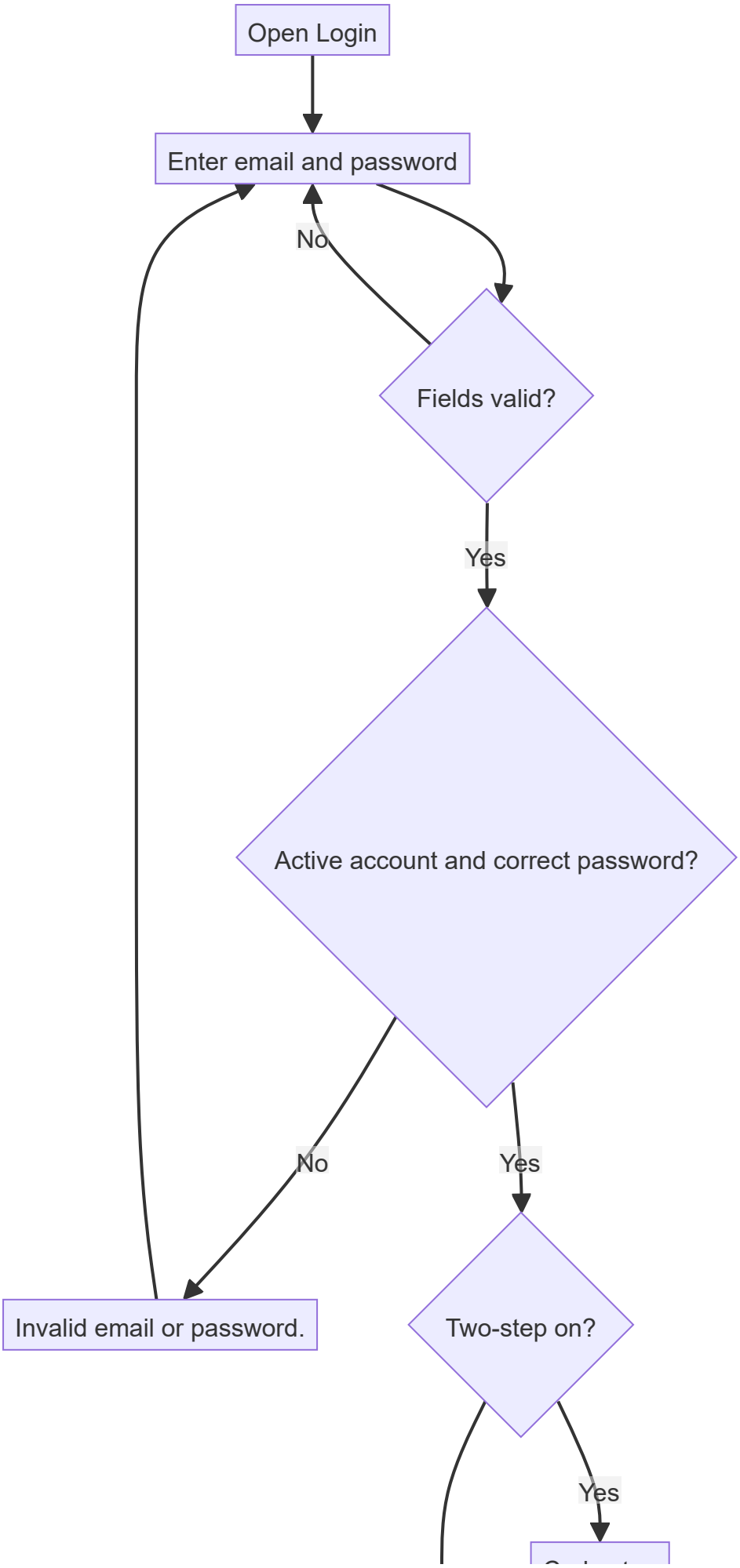
Functional requirements

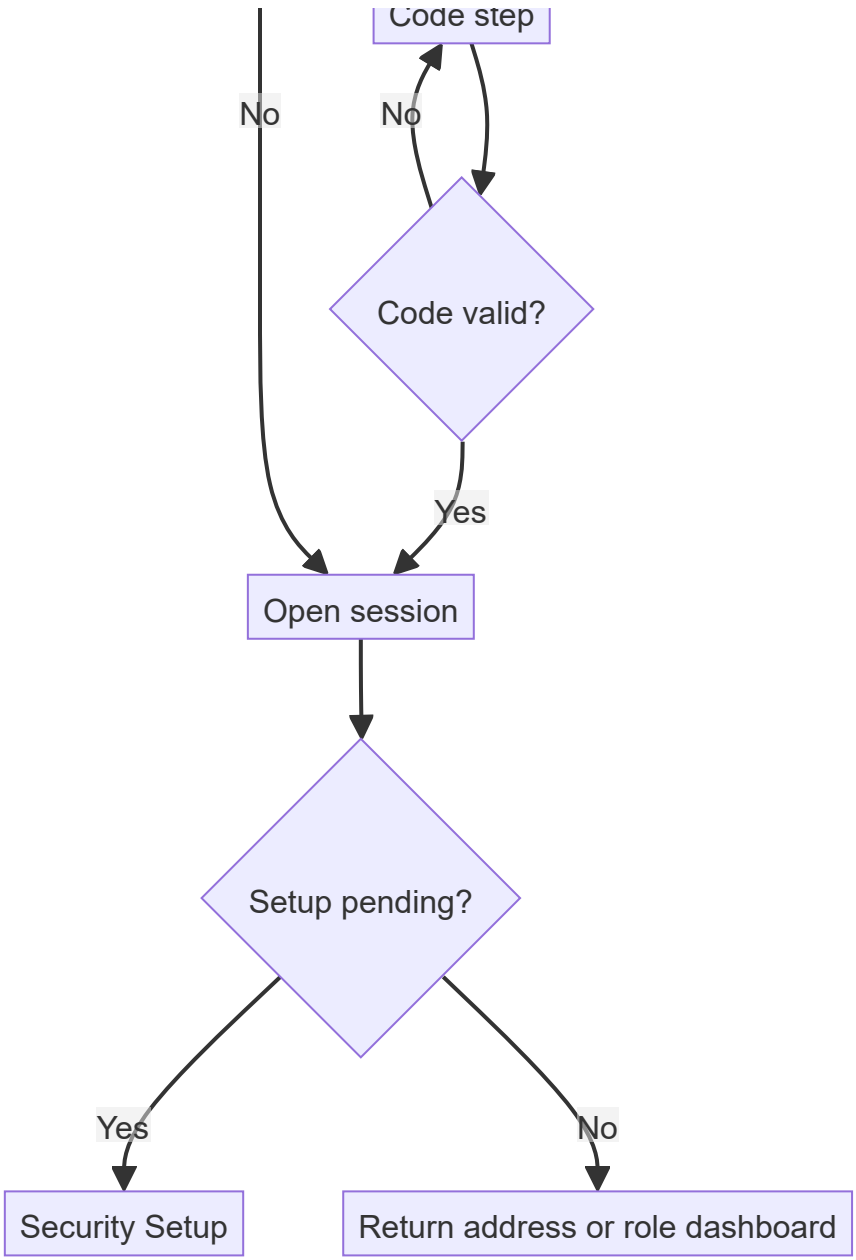
ID	Requirement	Priority	Permission	BRD trace
FR-AUTH-01	The system shall accept either the username or the email address of the account, together with the password, to sign in.	Must	Public	REQ-AUTH-01
FR-AUTH-02	The system shall require **Email** and **Password**, check the email format on the screen, and accept at most 256 characters for the identifier and 128 for the password on the server.	Must	Public	REQ-AUTH-01
FR-AUTH-03	The system shall answer an unknown account, an inactive or deleted account and a wrong password with the same message "Invalid email or password." and open no session.	Must	Public	REQ-AUTH-02
FR-AUTH-04	The system shall compare the password only against its stored salted hash and replace an old-format hash with the current format after the next successful sign-in.	Must	Public	REQ-AUTH-03
FR-AUTH-05	The system shall let the user show and hide the typed password with the eye button.	Should	Public	REQ-AUTH-11
FR-AUTH-06	The system shall pre-fill **Email** with the identifier last used for sign-in on the same device.	Could	Public	REQ-AUTH-12
FR-AUTH-07	The system shall record the date and time of each successful sign-in as the account's last login.	Should	Public	REQ-AUTH-10
FR-AUTH-08	The system shall, after a correct password, open the session directly when two-step verification is off, and otherwise show the Two-step verification step (4.1.2) without opening a session.	Must	Public	REQ-AUTH-04
FR-AUTH-09	The system shall, after sign-in, open the return address if it is a page inside TaskFlow; otherwise it shall open Administrator → `/admin`, Manager → `/manager`, custom roles → the widest dashboard their `dashboard.view.*` permissions allow, everyone else → `/me`.	Must	`dashboard.view.*` decides the target	REQ-AUTH-08
FR-AUTH-10	The system shall open Security Setup (`/security-setup`) instead of the dashboard when two-step verification is required but not yet set up.	Must	Public	REQ-AUTH-09

ID	Requirement	Priority	Permission	BRD trace
FR-AUTH-11	The system shall show in the notice area why the user was returned to sign in: session ended, credentials changed, or password reset.	Should	Public	REQ-AUTH-18, REQ-AUTH-29

Use case: UC-AUTH-01 – Sign in with password and authenticator code

Item	Detail
Actor	Any registered user (Administrator, Manager, Employee, Custom Role)
Preconditions	Account exists, is active and not deleted; two-step verification is on
Trigger	User enters email (or username) and password and clicks **Sign in**
Main flow	1. System checks the fields. 2. System finds the account and checks the password. 3. System creates a sign-in step valid for 5 minutes and shows **Two-step verification**. 4. User types the current 6-digit code and clicks **Verify**. 5. System checks the code (unused, within one 30-second step). 6. System opens the session, records the last login and opens the return address or the role dashboard.
Alternate flows	A1: two-step verification off and not required → session opens after step 2. A2: setup pending → session opens and Security Setup is shown (UC-AUTH-03). A3: ← **Back to sign in** → password step, password cleared.
Exceptions	E1: empty or invalid fields → field messages. E2: unknown, inactive, deleted account or wrong password → "Invalid email or password.". E3: wrong, reused or expired code → "The code is incorrect, or the sign-in has expired. Try again or sign in again.", code box cleared. E4: 5 wrong codes in a row → "Too many wrong codes. Try again in a few minutes." for 5 minutes.
Postconditions	Session open (access pass and renewal cookie); last login recorded; user on the right screen.


Open Login
Enter email and password
No
Fields valid?
Yes
Active account and correct password?
No
Yes
Invalid email or password.
Two-step on?
Yes



Field validations

Field	Rule	Error message
Email	Required	"Email is required."
Email	Valid email format (screen)	"Enter a valid email address."
Email	At most 256 characters (server)	Request refused as invalid
Password	Required	"Password is required."
Password	At most 128 characters (server)	Request refused as invalid

Business rules applied: BR-01, BR-02, BR-03, BR-28

Notifications / side effects

Event	Who is notified	Channel
Successful sign-in	Nobody; last login time recorded	–

Messages

Situation	Message
Wrong credentials, inactive or deleted account	"Invalid email or password."
Sent back after session ended or renewal failed	"Your session has ended, for example because your username or password was changed. Please sign in again."
Sent back after credentials changed	"Your username or password was changed. Sign in again with your new details."
Sent back after password reset	"Your password has been reset. Sign in with your new password."
No permission	Not applicable – public screen

4.1.2 Screen: Two-step verification (second step of Login)

Field	Value
Route / location	`/login`, shown after a correct password when two-step verification is on
Access	Public; needs the sign-in step created by FR-AUTH-08
Entry points	**Sign in** with a correct password

Screen elements

Element	Type	Description	Mandatory	Visible to
Two-step verification	Title	With "Enter the 6-digit code from your authenticator app."	–	Users with two-step verification on
Authentication code	Text input	Current code from the authenticator app	Yes	Same
Verify	Button	Submits the code	–	Same
← **Back to sign in**	Link	Returns to the password step	–	Same

Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-AUTH-12	The system shall accept the code step only within 5 minutes of the correct password, and refuse it earlier when the account's security stamp changed in the meantime (password change, password reset or MFA reset).	Must	Public	REQ-AUTH-05
FR-AUTH-13	The system shall accept a time-based 6-digit code from the current 30-second step or one step before or after it, and accept each code only once per account.	Must	Public	REQ-AUTH-06
FR-AUTH-14	The system shall lock code checking for the account for 5 minutes after 5 wrong codes in a row; the counter is shared with the code checks of My Profile.	Must	Public	REQ-AUTH-07
FR-AUTH-15	The system shall clear the code box after a refused code, and clear the password when the user returns with ← **Back to sign in**.	Should	Public	REQ-AUTH-04, REQ-AUTH-06

Use case: covered by UC-AUTH-01 (steps 3 – 6, exceptions E3 – E4).

Field validations

Field	Rule	Error message
Authentication code	Required	"Enter the code to continue."
Authentication code	Correct, unused, sign-in step not expired	"The code is incorrect, or the sign-in has expired. Try again or sign in again."
Authentication code	Not locked	"Too many wrong codes. Try again in a few minutes."

Business rules applied: BR-03, BR-28

Notifications / side effects

Event	Who is notified	Channel
Wrong code	Nobody; failure counted toward the 5-minute lockout	–

Messages

Situation	Message
Code refused without a server message	"That code did not work. Check the code and try again."
No permission	Not applicable – public step

> ⚠️ To be confirmed: the server also accepts an unused backup code at this step, but the code box takes only 6 characters while backup codes have the form `xxxxx-xxxxx`.

4.1.3 Screen: Forgot your password? (Account Recovery)

Field	Value
Route / location	`/account-recovery`
Access	Public; usable only for active accounts with two-step verification on
Entry points	**Forgot password?** on Login

Screen elements

Element	Type	Description	Mandatory	Visible to
Forgot your password?	Title	With a short explanation	–	Everyone
Email or username	Text input	Step 1 identifier	Yes	Everyone
Code from your authenticator app	Text input	Step 1 code, 6 digits	Yes	Everyone
Continue	Button	Verifies step 1	–	Everyone
Note	Text	"This works if you turned on two-step verification from your Profile. If you did not, ask an administrator to set a new password for you."	–	Everyone
"Code verified. Choose a new password for your account."	Subtitle	Step 2	–	Users who passed step 1

Element	Type	Description	Mandatory	Visible to
New password, **Confirm new password**	Password inputs	Step 2	Yes	Same
Reset password	Button	Saves the new password	–	Same
Start over	Button	Back to step 1	–	Only when the reset is refused
← **Back to sign in**	Link	Returns to Login	–	Everyone
Legal footer	Footer	4.2.2	–	Everyone

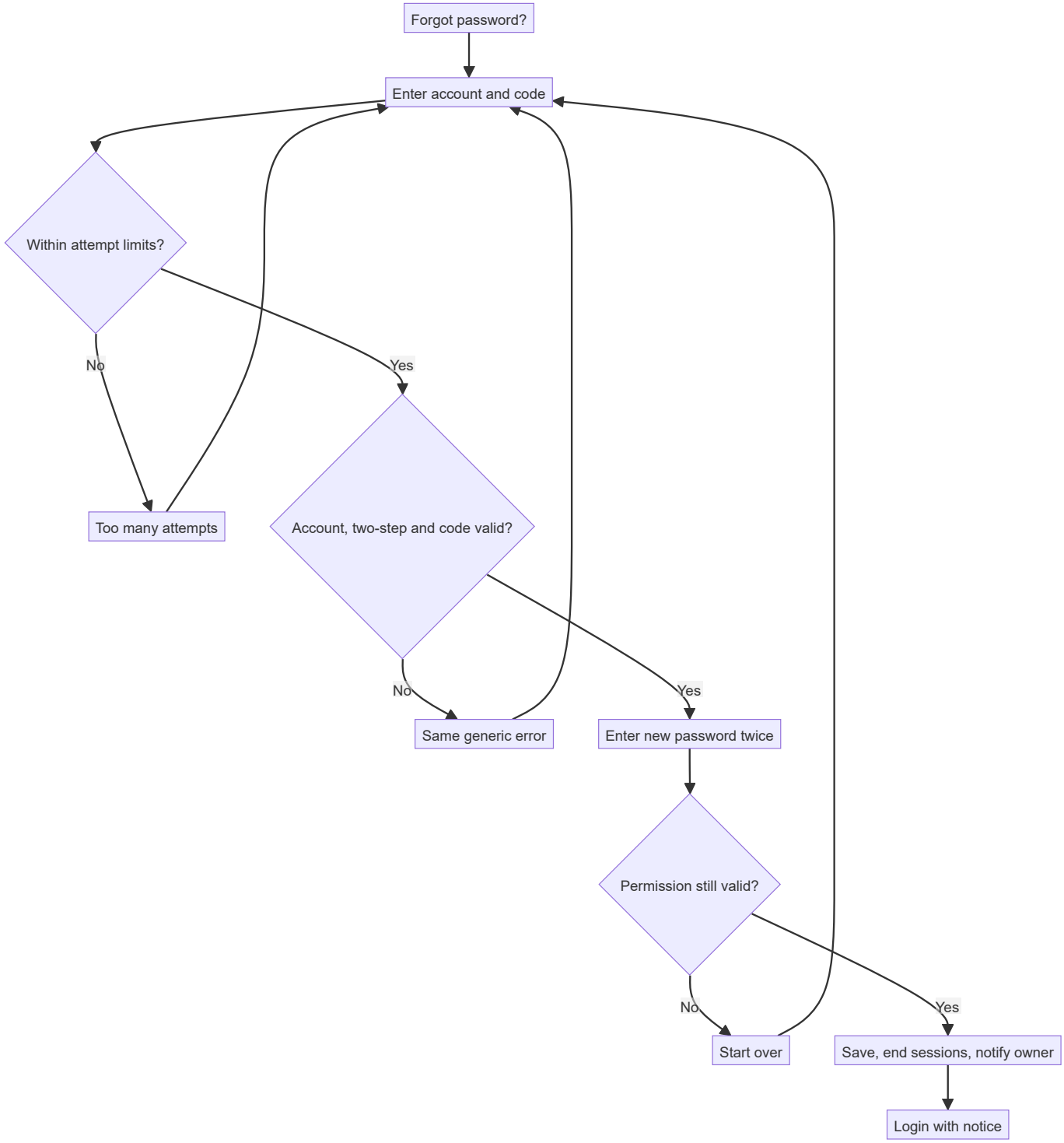
Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-AUTH-16	The system shall verify the identity with the email or username and the current 6-digit authenticator code.	Must	Public	REQ-AUTH-19
FR-AUTH-17	The system shall offer recovery only for active, not deleted accounts that have two-step verification on.	Must	Public	REQ-AUTH-20
FR-AUTH-18	The system shall refuse a backup code as the recovery factor.	Must	Public	REQ-AUTH-21
FR-AUTH-19	The system shall show the same message for an unknown account, an inactive account, an account without two-step verification and a wrong code.	Must	Public	REQ-AUTH-22
FR-AUTH-20	The system shall allow at most 5 verification attempts per account and 20 per network address in any sliding 5 minutes and refuse further attempts.	Must	Public	REQ-AUTH-23
FR-AUTH-21	The system shall not count wrong recovery codes toward the sign-in code lockout, and shall still refuse an account that is currently locked for codes.	Must	Public	REQ-AUTH-24
FR-AUTH-22	The system shall grant a reset permission valid for 10 minutes and usable once; when it is used, all other open reset permissions of the user are cancelled; when only the new password is rejected, the same permission stays usable until it expires.	Must	Public	REQ-AUTH-25
FR-AUTH-23	The system shall require a new password of 6 to 128 characters entered identically in **New password** and **Confirm new password**.	Must	Public	REQ-AUTH-26
FR-AUTH-24	The system shall end every session of the user on all devices when the password is reset.	Must	Public	REQ-AUTH-27
FR-AUTH-25	The system shall send the account owner the System notification "Your password was reset with your authenticator app. If this was not you, contact an administrator now." after a successful reset; a failure to write it does not undo the reset.	Must	Public	REQ-AUTH-28
FR-AUTH-26	The system shall open Login with the notice "Your password has been reset. Sign in with your new password." after the reset.	Should	Public	REQ-AUTH-29

ID	Requirement	Priority	Permission	BRD trace
FR-AUTH-27	The system shall show **Start over** when the reset permission has expired or was already used, returning to step 1.	Should	Public	REQ-AUTH-30

Use case: UC-AUTH-02 – Reset a forgotten password with the authenticator app

Item	Detail
Actor	User whose two-step verification is on
Preconditions	Not signed in; authenticator app at hand
Trigger	Clicks **Forgot password?** on Login
Main flow	1. User enters email or username and the current code, clicks **Continue**. 2. System checks attempt limits, account, two-step status and code. 3. System grants a 10-minute single-use reset permission and shows step 2. 4. User enters and confirms a new password, clicks **Reset password**. 5. System saves the password, ends all sessions, cancels other reset permissions, sends the System notification. 6. System opens Login with the success notice.
Alternate flows	A1: ← **Back to sign in** at any time. A2: new password rejected by validation → user corrects it and retries with the same permission.
Exceptions	E1: limits exceeded → "Too many attempts. Try again in a few minutes.". E2: any verification failure → "The details are incorrect, or the code has expired. Check them and try again.". E3: permission expired or used → "This reset link is no longer valid. Start again from the login page." and **Start over**.
Post conditions	New password active; every old session ended; owner notified.



Field validations

Field	Rule	Error message
Email or username	Required; at most 256 characters	"Email or username is required."
Code from your authenticator app	Required; exactly 6 digits on the screen (server accepts up to 32 characters)	"Enter the 6-digit code from the app."
New password	6 – 128 characters	"Use at least 6 and at most 128 characters."
Confirm new password	Same as New password	"Passwords do not match."

Business rules applied: BR-02, BR-03, BR-23, BR-25, BR-28

Notifications / side effects

Event	Who is notified	Channel
Password reset with authenticator	Account owner	In-app (System) – catalogue N-12
Password reset	All sessions of the user end	–

Messages

Situation	Message
Verification failed	"The details are incorrect, or the code has expired. Check them and try again."
Too many attempts	"Too many attempts. Try again in a few minutes."
Verification failed without a server message	"Could not verify the code. Please try again."
Reset permission used or expired	"This reset link is no longer valid. Start again from the login page."
Reset failed without a server message	"Could not reset your password. Please try again."
Success (on Login)	"Your password has been reset. Sign in with your new password."
No permission	Not applicable – public screen

4.1.4 Screen: Let's secure your account (Security Setup)

Field	Value
Route / location	`/security-setup`
Access	Signed-in users with setup pending; `profile.view`, `profile.edit`, `profile.mfa.manage` (Always On). Users without setup pending are sent to their dashboard; visitors to Login
Entry points	Automatic after sign-in (FR-AUTH-10); any other screen while setup is pending

Screen elements

Element	Type	Description	Mandatory	Visible to
Progress list **Your details**, **Authenticator app**, **Backup codes**	Stepper	"Step N of 3" on small screens	–	Users with setup pending
"Welcome, (first name)" and Employee code, Username, Role, Projects, Managers	Read-only facts	Step 1	–	Same
Full name	Text input	Step 1, at most 200 characters	Yes	Same
Email	Text input with fixed `@taskflow.com`	Step 1, name part only	Yes	Same
Mobile number	Text input	Step 1, international format	Yes	Same
Sign out, **Save and continue**	Buttons	Step 1	–	Same
Current password	Password input with **Show/Hide**	Step 2	Yes	Same
Back, **Show my QR code**	Buttons	Step 2	–	Same
QR code, typed key, **Copy key**	Display, button	Step 2 after the password check	–	Same

Element	Type	Description	Mandatory	Visible to
6-digit code	Text input	Step 2	Yes	Same
Start over, **Verify and turn on**	Buttons	Step 2	–	Same
"Two-step verification is on", 10 backup codes, **Copy all**, **Download**	Display, buttons	Step 3	–	Same
"I have saved my backup codes somewhere safe."	Check box	Step 3	Yes	Same
Open TaskFlow	Button	Enabled only after the check box	–	Same
"We could not load your details." with **Try again**	Error box	Load failure	–	Same

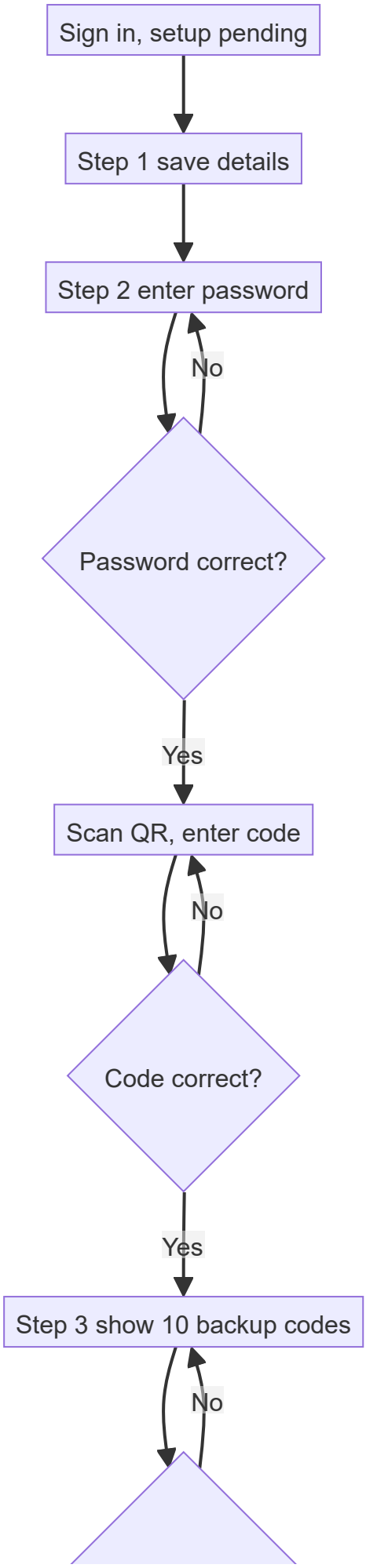
Functional requirements

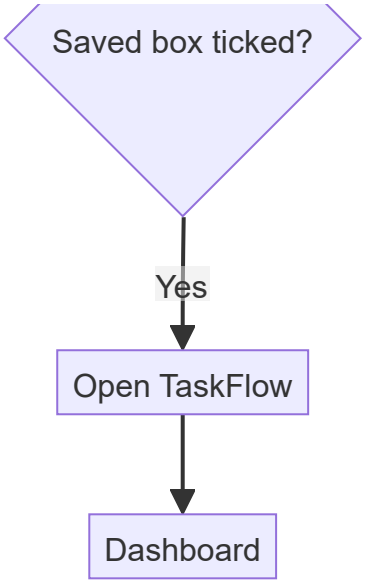
ID	Requirement	Priority	Permission	BRD trace
FR-AUTH-28	The system shall require two-step verification for every user by default; it can be switched off only in server configuration as a break-glass measure.	Must	–	REQ-AUTH-31
FR-AUTH-29	The system shall, while setup is pending, open only `/security-setup` and refuse every server function except sign-in, the user's own profile and the MFA setup functions with "Set up two-step verification to continue.".	Must	–	REQ-AUTH-32
FR-AUTH-30	The system shall send a signed-in user without setup pending who opens `/security-setup` to their dashboard, and a visitor to Login.	Must	–	REQ-AUTH-32
FR-AUTH-31	The system shall show the user's read-only facts and let the user confirm and save **Full name**, **Email** and **Mobile number** before continuing.	Must	`profile.view`, `profile.edit`	REQ-AUTH-33
FR-AUTH-32	The system shall show a new setup key and QR code only after the current password is checked, allowing 5 password attempts per account in 5 minutes; two-step verification stays off at this point.	Must	`profile.mfa.manage`	REQ-AUTH-34
FR-AUTH-33	The system shall let the user copy the setup key when the QR code cannot be scanned.	Should	`profile.mfa.manage`	REQ-AUTH-35
FR-AUTH-34	The system shall turn on two-step verification only after a correct 6-digit code from the app for the key just issued; a backup code is not accepted; wrong codes count toward the shared 5-minute lockout.	Must	`profile.mfa.manage`	REQ-AUTH-36
FR-AUTH-35	The system shall refuse to start setup again while two-step verification is on.	Must	`profile.mfa.manage`	REQ-AUTH-37
FR-AUTH-36	The system shall create exactly 10 single-use backup codes in the form `XXXXX-XXXXX` when two-step verification is turned on and show them only once, with **Copy all** and **Download**.	Must	`profile.mfa.manage`	REQ-AUTH-38
FR-AUTH-37	The system shall keep **Open TaskFlow** disabled until the check box is ticked, then open the user's dashboard.	Must	–	REQ-AUTH-39

ID	Requirement	Priority	Permission	BRD trace
FR-AUTH-38	The system shall store the setup key only encrypted and the backup codes only as hashes, and never show them again.	Must	–	REQ-AUTH-40
FR-AUTH-39	The system shall let the user leave with **Sign out** without finishing setup.	Should	–	REQ-AUTH-41
FR-AUTH-40	The system shall show "We could not load your details." with **Try again** when the details cannot be loaded.	Should	`profile.view`	REQ-AUTH-33

Use case: UC-AUTH-03 – Complete first-time Security Setup

Item	Detail
Actor	Signed-in user with setup pending
Preconditions	Signed in (UC-AUTH-01 A2); authenticator app installed on the phone
Trigger	System opens **Let's secure your account**
Main flow	1. User checks Full name, Email and Mobile number, clicks **Save and continue**. 2. User enters **Current password**, clicks **Show my QR code**. 3. System checks the password and shows QR code and key. 4. User scans the QR code (or copies the key), enters the 6-digit code, clicks **Verify and turn on**. 5. System checks the code, turns two-step verification on and shows 10 backup codes. 6. User copies or downloads the codes, ticks the check box, clicks **Open TaskFlow**. 7. System opens the user's dashboard.
Alternate flows	A1: **Sign out** at step 1. A2: **Start over** at step 2 gets a new key. A3: **Back** at step 2 returns to step 1.
Exceptions	E1: invalid details → field messages. E2: wrong password → "Current password is incorrect."; 6th attempt in 5 minutes → "Too many attempts. Try again in a few minutes.". E3: wrong code → "The code is incorrect or has expired."; 5 wrong codes → lockout. E4: already on → "Authenticator MFA is already on. Turn it off first to set it up again.".
Postconditions	Two-step verification on; 10 backup codes issued; every later sign-in asks for a code.


Sign in, setup pending
Step 1 save details
Step 2 enter password
No
Password correct?
Yes
Scan QR, enter code
No
Code correct?
Yes
Step 3 show 10 backup codes
No



Field validations

Field	Rule	Error message
Full name	Required; at most 200 characters	"Enter your full name."
Email	Name part with letters, digits, dot, underscore or hyphen, then `@taskflow.com`	"Use letters, digits, dot, underscore or hyphen before @taskflow.com." (server: "Email must be a name using letters, digits, dot, underscore or hyphen followed by @taskflow.com.")
Email	Unique	"Email already exists."
Email + Mobile number	Not the same pair as another user	"A user with this email and mobile number already exists."
Mobile number	Required	"Enter your mobile number."
Mobile number	International format	"Use international format, for example +919876543210."
Current password	Required / correct	"Enter your password." / "Current password is incorrect."
6-digit code	Exactly 6 digits	"Enter the 6-digit code from the app."
6-digit code	Correct and setup started	"The code is incorrect or has expired." / "Start the setup first to get a key."

Business rules applied: BR-03, BR-24, BR-28

Notifications / side effects

Event	Who is notified	Channel
Two-step verification turned on	Nobody; 10 backup codes created	–

Messages

Situation	Message
Details not saved without a server message	"Could not save your details. Please try again."

Situation	Message
Setup failed without a server message	"Could not start setup. Please try again."
Code failed without a server message	"That code did not work. Check the app and try the new code."
Too many password attempts / wrong codes	"Too many attempts. Try again in a few minutes." / "Too many wrong codes. Try again in a few minutes."
Copy done / failed	"Key copied" / "Backup codes copied" / "Could not copy. Select the text and copy it by hand."
Any other page while setup pending	"Set up two-step verification to continue."

> ⚠️ To be confirmed: regenerating backup codes exists as a server function (REQ-PROF-29) but no screen offers it.

4.1.5 Screen: Sessions and Sign out (all signed-in screens)

Field	Value
Route / location	**Sign out** icon in the header account area of every signed-in screen (and on Security Setup step 1); background session handling
Access	All signed-in users; no permission key
Entry points	Successful sign-in (FR-AUTH-08)

Screen elements

Element	Type	Description	Mandatory	Visible to
Sign out	Icon button	Ends the session on this device	–	All signed-in users
Login notice	Info box	Shown after the session ends (4.1.1)	–	Users returned to Login

Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-AUTH-41	The system shall issue at sign-in a 30-minute access pass and a 7-day renewal cookie (HttpOnly, Secure, SameSite Strict, sent only to the sign-in functions) and restore the session from the cookie after a page reload.	Must	–	REQ-AUTH-13
FR-AUTH-42	The system shall, when a request is refused because the access pass expired, renew the pass silently with the cookie and repeat the request.	Must	–	REQ-AUTH-13
FR-AUTH-43	The system shall replace the renewal cookie at every renewal and refuse a used, expired or revoked cookie.	Must	–	REQ-AUTH-14
FR-AUTH-44	The system shall check the session every 15 seconds and when the browser tab becomes visible again, and apply the user's current role and permissions to menus and screens without a new sign-in.	Must	–	REQ-AUTH-15
FR-AUTH-45	The system shall, on every request, refuse a session whose user no longer exists, is inactive or deleted, or whose security stamp has changed, and read the role and permissions afresh.	Must	–	REQ-AUTH-16

ID	Requirement	Priority	Permission	BRD trace
FR-AUTH-46	The system shall, on **Sign out**, revoke and remove the renewal cookie and open Login; signing out without a cookie or with an already revoked one still succeeds quietly.	Must	–	REQ-AUTH-17
FR-AUTH-47	The system shall, when renewal fails, clear the session and open Login with "Your session has ended, for example because your username or password was changed. Please sign in again.".	Should	–	REQ-AUTH-18
FR-AUTH-48	The system shall change the security stamp and revoke all renewal cookies of the user after a password change, a password reset and an administrator MFA reset, so every session on every device ends.	Must	–	REQ-AUTH-16, REQ-AUTH-27
FR-AUTH-49	The system shall check the permission key of every protected function on the server, independently of the screen, and refuse a missing permission with "403 Forbidden".	Must	Per function (3.2)	REQ-AUTH-16
FR-AUTH-50	The system shall show in the menu only the entries the user's permissions allow and send a user who opens a screen without its permission to their own dashboard.	Must	Per screen (2.4)	REQ-AUTH-15
FR-AUTH-51	The system shall keep only Login, Account Recovery and the legal pages public, send a visitor who opens a protected screen to Login with the address as return address, and send any unknown address to Login.	Must	Public	REQ-AUTH-08

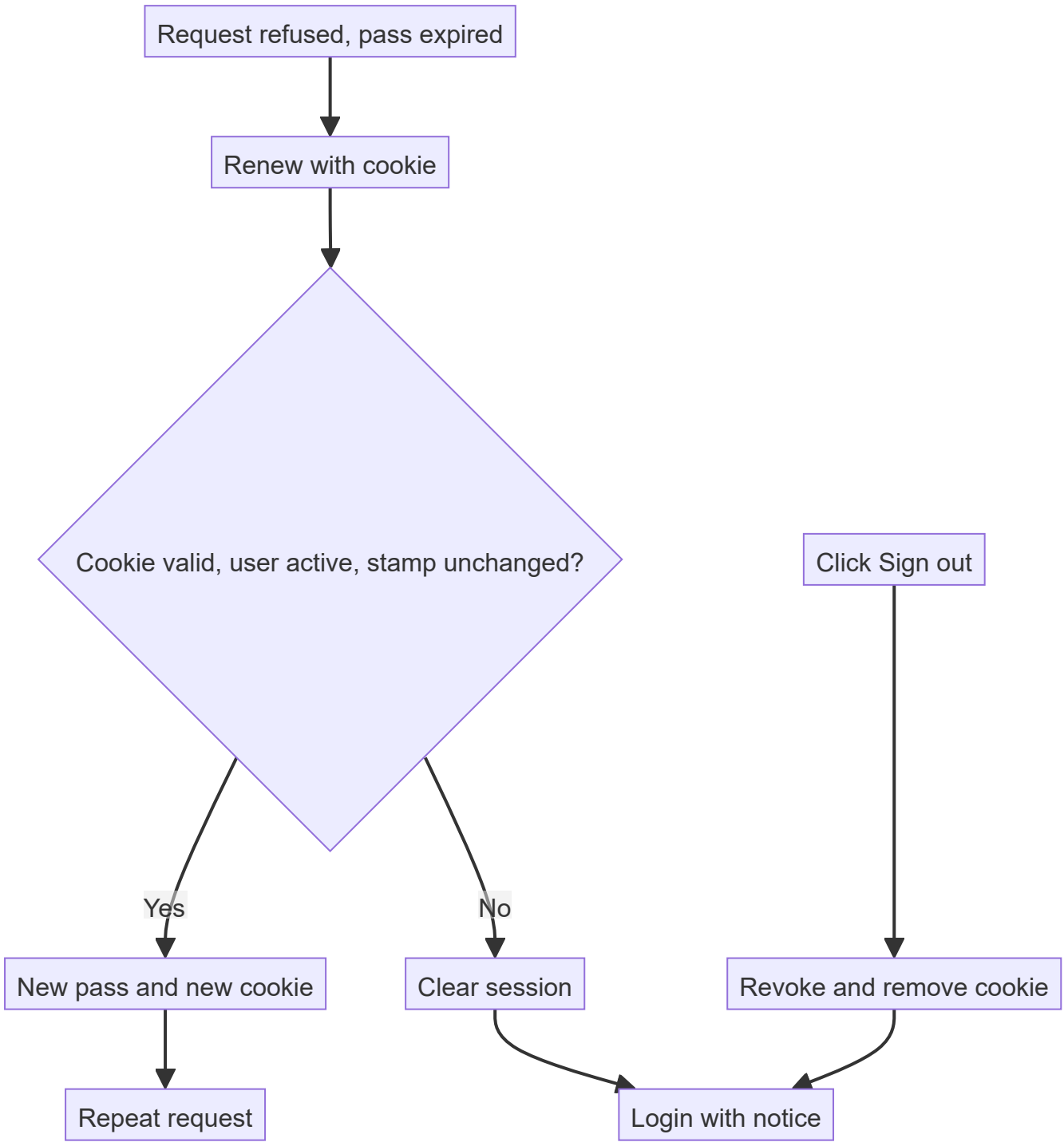
Use case: UC-AUTH-04 – Keep the session alive and apply changes

Item	Detail
Actor	Signed-in user; System
Preconditions	Session open
Trigger	Access pass expires, 15-second check, tab becomes visible, or a security change happens
Main flow	1. A request is refused because the pass expired. 2. System renews with the cookie, issuing a new pass and a new cookie. 3. System repeats the request. 4. Every 15 seconds the system re-reads the user, role and permissions and updates menus.
Alternate flows	A1: an administrator changed the user's role or a custom role's permissions → the next check applies them (custom-role permission change signs holders out, REQ-ROLE-17).
Exceptions	E1: cookie expired, revoked or reused, user inactive or deleted, security stamp changed → session cleared, Login with notice. E2: setup pending detected → Security Setup.
Postconditions	Session continues with current rights, or ends safely.

Use case: UC-AUTH-05 – Sign out

Item	Detail
Actor	Signed-in user
Preconditions	Session open (or setup pending on Security Setup step 1)
Trigger	Clicks **Sign out**
Main flow	1. System revokes and removes the renewal cookie. 2. System clears the session on the device. 3. System opens Login.
Alternate flows	A1: no cookie or already revoked → still signed out quietly.

Item	Detail
Exceptions	None
Postconditions	No usable session on this device.



Field validations

Field	Rule	Error message
–	No input fields	*Not applicable for this screen.*

Business rules applied: BR-01, BR-02, BR-04, BR-05, BR-06, BR-25

Notifications / side effects

Event	Who is notified	Channel
Password change, password reset, MFA reset	All sessions of the user end	–
Account deactivated or deleted (FRD-Employee-Management)	Session refused on next request	–

Messages

Situation	Message
Session ended or renewal failed	"Your session has ended, for example because your username or password was changed. Please sign in again."
Credentials changed	"Your username or password was changed. Sign in again with your new details."
Setup pending (stale session)	"Set up two-step verification to continue."
Concurrent change	"Someone else changed this record while you were saving. Reload it and try again."
Unexpected error	"Something went wrong. Please try again."
No permission (screen)	User returned to their dashboard; no message
No permission (function)	"403 Forbidden"; the screen shows its fallback error

4.2 Module: Legal (LEGAL)

Purpose: Publish TaskFlow's five legal documents to everyone and show the same ownership statement on every main screen. · **BRD reference:** Overall BRD SCREEN 4 – 5, REQ-LEGAL-01 – REQ-LEGAL-15, BR-01, BR-29

4.2.1 Screen: Legal Document Page (Copyright / Terms of Use / Privacy Policy / Disclaimer / Licences)

Field	Value
Route / location	`/legal/copyright`, `/legal/terms`, `/legal/privacy`, `/legal/disclaimer`, `/legal/licenses`
Access	Public – Visitor, Administrator, Manager, Employee, Custom Role; no permission key
Entry points	Legal footer links (4.2.2); typing the address. No side-menu entry

Screen elements

Element	Type	Description	Mandatory	Visible to
Top bar	Header	"TF" mark and "TaskFlow — Intelligent Task Allocation & Tracking System"	–	Everyone
Back to the application / **Back to sign in**	Link	Depends on whether the user is signed in	–	Everyone
Heading	Text	Title, one-line summary, "Last updated: 21 October 2026 · © 2026 Jojo George. All Rights Reserved."	–	Everyone
Contents	Link list	Numbered sections (Copyright 8, Terms 14, Privacy 11, Disclaimer 7, Licences 3)	–	Everyone
Document body	Text, lists, notes, tables	"What we collect" table (Privacy), "Third-party components" table (Licences)	–	Everyone
Ownership line	Text	At the foot of the page	–	Everyone

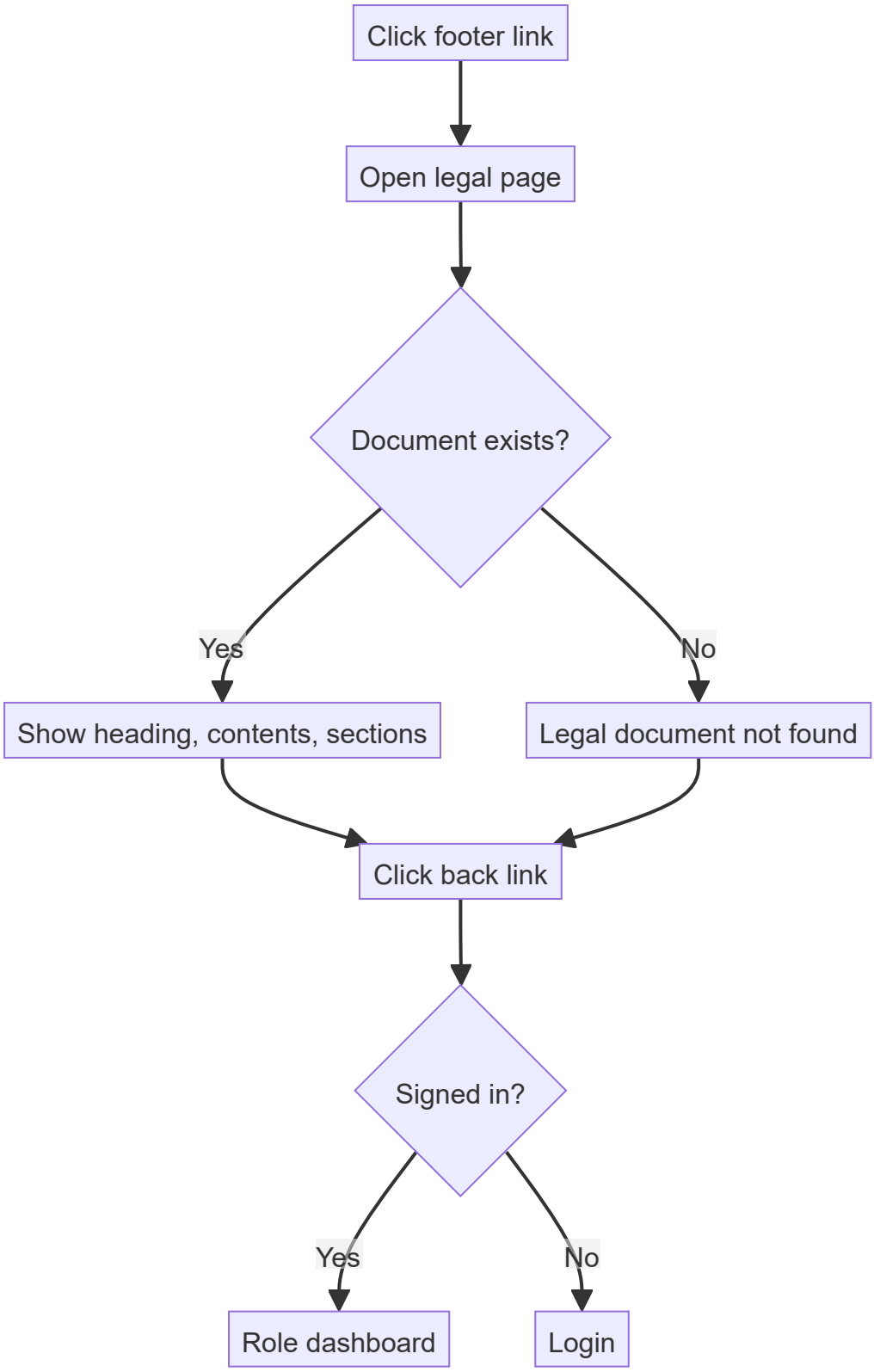
Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-LEGAL-01	The system shall publish exactly five legal documents, each on its own page: Copyright Notice (`/legal/copyright`), Terms of Use (`/legal/terms`), Privacy Policy (`/legal/privacy`), Disclaimer (`/legal/disclaimer`) and Third-Party Licence Notices (`/legal/licenses`).	Must	Public	REQ-LEGAL-01
FR-LEGAL-02	The system shall open the legal pages without sign-in, without any permission and also while two-step setup is pending.	Must	Public	REQ-LEGAL-02
FR-LEGAL-03	The system shall show on each page the title, summary, "Last updated" date and the ownership line in the heading and at the foot.	Must	Public	REQ-LEGAL-03
FR-LEGAL-04	The system shall show a **Contents** list of the document's sections and scroll to a section when its entry is clicked.	Must	Public	REQ-LEGAL-03
FR-LEGAL-05	The system shall insert owner name (Jojo George), year (2026), contact email (jojogeorge3344@gmail.com), rights statement ("All Rights Reserved") and application name from the single owner record into every document.	Must	Public	REQ-LEGAL-04
FR-LEGAL-06	The system shall label the back link **Back to the application** for a signed-in user and send Administrator → `/admin` , Manager → `/manager` , any other role → `/me` ; for a visitor it shall label it **Back to sign in** and open `/login` .	Should	Public	REQ-LEGAL-05
FR-LEGAL-07	The system shall show "Legal document not found" with "The document you requested is not available." when no document matches the page.	Should	Public	REQ-LEGAL-06
FR-LEGAL-08	The system shall keep the documents read-only: no edit function exists and opening a page creates no record, notification, mail or activity entry.	Must	Public	REQ-LEGAL-07
FR-LEGAL-09	The Privacy Policy shall state what data TaskFlow keeps and why, where it is stored (data folder on the server, default `C:\TaskFlow` on Windows), who can see it, how it is protected (salted password hash, encrypted AI keys) and how long it is kept.	Must	Public	REQ-LEGAL-08
FR-LEGAL-10	The Terms of Use and the Privacy Policy shall state that, when a cloud AI model is chosen, the question and the task details behind it are sent to that provider, and that a notice is shown the first time a cloud model is used.	Must	Public	REQ-LEGAL-09
FR-LEGAL-11	The Disclaimer shall state that assistant output and allocation suggestions are advisory only and must be checked before acting on them.	Must	Public	REQ-LEGAL-10

Use case: UC-LEGAL-01 – Read a legal document

Item	Detail
Actor	Anyone (Visitor or any signed-in role)

Item	Detail
Preconditions	None
Trigger	Clicks a footer link or opens a `/legal/...` address
Main flow	1. System opens the page without checking sign-in. 2. System fills the owner values into the text. 3. System shows heading, **Contents** and all sections. 4. User clicks a **Contents** entry; the page scrolls to that section. 5. User clicks the back link and returns to the dashboard (signed in) or Login (visitor).
Alternate flows	A1: user leaves through a link in the text or the browser back button.
Exceptions	E1: no matching document → "Legal document not found".
Postconditions	Nothing stored.



Field validations

Field	Rule	Error message
–	Read-only page, no input	*Not applicable for this screen.*

Business rules applied: BR-01, BR-29

Notifications / side effects

Event	Who is notified	Channel
Page opened	Nobody; nothing recorded	–

Messages

Situation	Message
Document not found	"Legal document not found" / "The document you requested is not available." / "Choose one of the documents from the footer, or back to sign in." ("back to the application" when signed in)
Back link	"Back to the application" (signed in) / "Back to sign in" (not signed in)
No permission	Not applicable – public page

> ⚠️ To be confirmed: an address under `/legal/` that is not one of the five documents is currently sent to Login by the catch-all route, so the "not found" message may not be reached.

4.2.2 Screen: Legal Footer and Ownership Notice (part of other screens)

Field	Value
Route / location	Bottom of `/login`, `/account-recovery` (dark style) and every signed-in screen (light style)
Access	Public – everyone
Entry points	Always visible on those screens

Screen elements

Element	Type	Description	Mandatory	Visible to
Ownership line	Text	"© 2026 Jojo George. All Rights Reserved. · TaskFlow — Intelligent Task Allocation & Tracking System."	–	Everyone
Copyright, **Terms of Use**, **Privacy Policy**, **Disclaimer**, **Licences**	Links	Open the matching document in the same tab	–	Everyone
Browser page details	Page title and details	Title "TaskFlow — Intelligent Task Allocation & Tracking System", author, copyright, description	–	Everyone (browser tab)

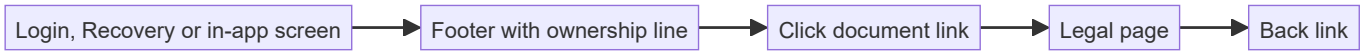
Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-LEGAL-12	The system shall show the legal footer with the ownership line at the bottom of Login, Account Recovery and every signed-in screen.	Must	Public	REQ-LEGAL-11
FR-LEGAL-13	The system shall show the five document links in the fixed order Copyright, Terms of Use, Privacy Policy, Disclaimer, Licences, each opening its page in the same tab.	Must	Public	REQ-LEGAL-12
FR-LEGAL-14	The system shall take the footer's ownership values from the same owner record as the documents, so the statement is identical everywhere.	Must	Public	REQ-LEGAL-13

ID	Requirement	Priority	Permission	BRD trace
FR-LEGAL-15	The system shall not show the footer on the legal pages, which carry their own ownership line.	Could	Public	REQ-LEGAL-14
FR-LEGAL-16	The system shall set, when the application starts, the browser page title "TaskFlow — Intelligent Task Allocation & Tracking System" and the author "Jojo George", the copyright "© 2026 Jojo George. All Rights Reserved." and a one-line description.	Should	Public	REQ-LEGAL-15

Use case: UC-LEGAL-02 – Open a legal document from the footer

Item	Detail
Actor	Anyone
Preconditions	User is on Login, Account Recovery or any signed-in screen
Trigger	Clicks a footer link, for example **Licences**
Main flow	1. System opens the matching page (`/legal/licenses`) in the same tab. 2. UC-LEGAL-01 continues.
Alternate flows	None
Exceptions	None
Postconditions	Legal page shown; nothing stored.



Field validations

Field	Rule	Error message
–	No input	*Not applicable for this screen.*

Business rules applied: BR-29

Notifications / side effects

Event	Who is notified	Channel
–	Nobody	–

Messages

Situation	Message
Ownership line	"© 2026 Jojo George. All Rights Reserved. · TaskFlow — Intelligent Task Allocation & Tracking System."
Validation / no permission	Not applicable – read-only, public

4.3 Other Modules (summary – detail in the sub-module FRDs)

Code	Main functional areas	Key global rules	FRD
DASH	Role dashboards, analytics, filters (default Last 30 days by Created date), drill-down to exactly the counted tasks	BR-06, BR-12, BR-17, BR-22	FRD-Dashboard

Code	Main functional areas	Key global rules	FRD
TASK	Task List, Create/Copy, Details, edit, status, worked hours, recipients, reassign with reason, soft delete, comments, activity history	BR-07 – BR-12, BR-18, BR-19, BR-21	FRD-Task-Management
WORK	Active, high-priority, overdue, completed counts and allocation per employee; "High load" at twice the average	BR-06, BR-12	FRD-Employee-Workload
BOARD	Five status columns, drag and drop under the same status rules, filters	BR-10, BR-11	FRD-Board
EMP	Employee list, add/edit, activate/deactivate, delete protections, role assignment, **Reset MFA**	BR-02, BR-21, BR-24 – BR-26	FRD-Employee-Management
PROJ	Projects, team members, positions, allocation percentages (at most 100% per person), project manager	BR-21	FRD-Project-Management
DEPT	Departments, managers and members; no delete while members exist	BR-21	FRD-Department-Management
MAIL	Private mailbox, folders, compose/reply/forward, drafts, attachments, system mail from "TaskFlow"	BR-20, BR-23	FRD-Mail
NOTIF	Notification Center and bell; event notifications; reminder engine every 60 seconds	BR-13 – BR-17, BR-20, BR-23	FRD-Notification
RPT	Task totals and completion summary with filters; CSV export up to 50,000 rows	BR-06, BR-22	FRD-Reports
REM	Reminders on/off, deadline-day reminder, default first/second reminder, overdue frequency, Mail delivery	BR-14	FRD-Reminder-Settings
APP	Application name, timezone, default priority, in-app notifications	BR-17	FRD-Application-Settings
TFAS	AI provider, model, server address, encrypted keys, connection test	BR-28	FRD-TFA-Settings
ROLE	Roles list, permission tree, compare view, custom roles, prerequisites	BR-04, BR-05	FRD-Role-Configuration
PROF	Own details, change password with code, self-service two-step verification	BR-03, BR-24, BR-25	FRD-My-Profile
TFA	Chat panel, help topics, Action Card with confirm, audit log, rate limit, voice input	BR-20, BR-27, BR-28	FRD-Assistant

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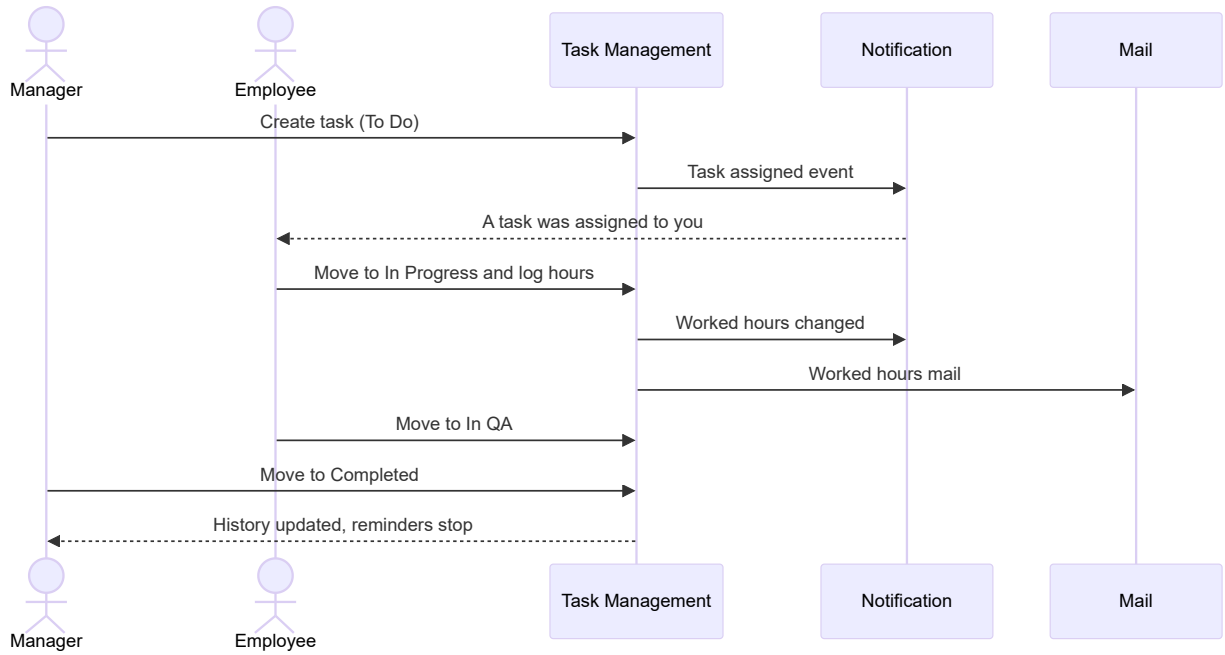
5. Cross-Module Workflows

In the FR reference column, sub-module requirements are named by their FRD and the traced BRD requirement (`FR-<CODE>` ← `REQ-<CODE>-nn`); the exact `FR-<CODE>-nn` number is in that FRD's traceability matrix.

5.1 Task Lifecycle (create → work → review → complete)

Step	Module	Actor / System	Action	FR reference
1	AUTH	Manager	Signs in and opens Create Task	FR-AUTH-01, FR-AUTH-09
2	TASK	Manager	Creates the task (To Do, code `TSK-` + 6 digits, assignee as recipient, reminder days prefilled from Reminder Settings, priority from Application Settings)	FR-TASK ← REQ-TASK-12, REQ-TASK-19 – REQ-TASK-23; FR-REM ← REQ-REM-15; FR-APP ← REQ-APP-10

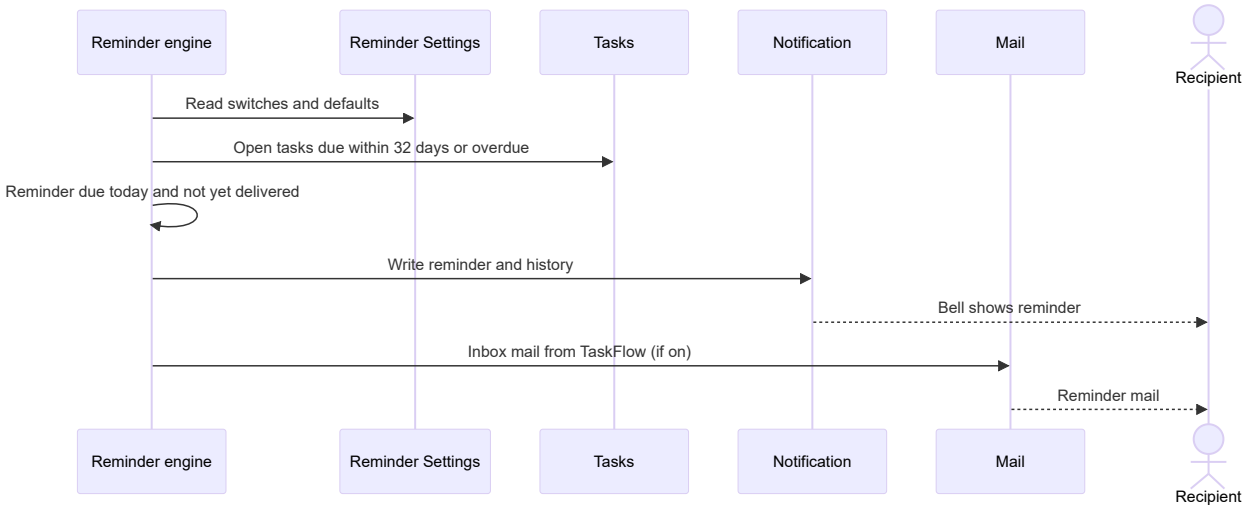
Step	Module	Actor / System	Action	FR reference
3	NOTIF	System	Notifies the assignee "A task was assigned to you: …" (N-01)	FR-NOTIF ← REQ-NOTIF-20; FR-TASK ← REQ-TASK-24
4	BOARD, TASK	Employee	Moves the task to In Progress and logs worked hours	FR-BOARD ← REQ-BOARD-07, REQ-BOARD-08; FR-TASK ← REQ-TASK-32
5	NOTIF, MAIL	System	Notifies assignee and recipients of the hours change (N-05, M-04)	FR-TASK ← REQ-TASK-34; FR-NOTIF ← REQ-NOTIF-23; FR-MAIL ← REQ-MAIL-25
6	TASK, NOTIF, MAIL	Employee / Manager	Discuss in comments; members notified (N-04, M-02)	FR-TASK ← REQ-TASK-51; FR-NOTIF ← REQ-NOTIF-22; FR-MAIL ← REQ-MAIL-23
7	TASK	Manager	Reassigns with a reason if needed (N-02, N-03, M-03)	FR-TASK ← REQ-TASK-41 – REQ-TASK-44
8	TASK, BOARD	Employee / Manager	Moves to In QA, then Completed; history records each move; reminders stop	FR-BOARD ← REQ-BOARD-11; FR-TASK ← REQ-TASK-52; FR-NOTIF ← REQ-NOTIF-30
9	DASH, RPT	System	Dashboards and reports count the task with the same rules	FR-DASH ← REQ-DASH-27; FR-RPT ← REQ-RPT-*



5.2 Automatic Deadline Reminders

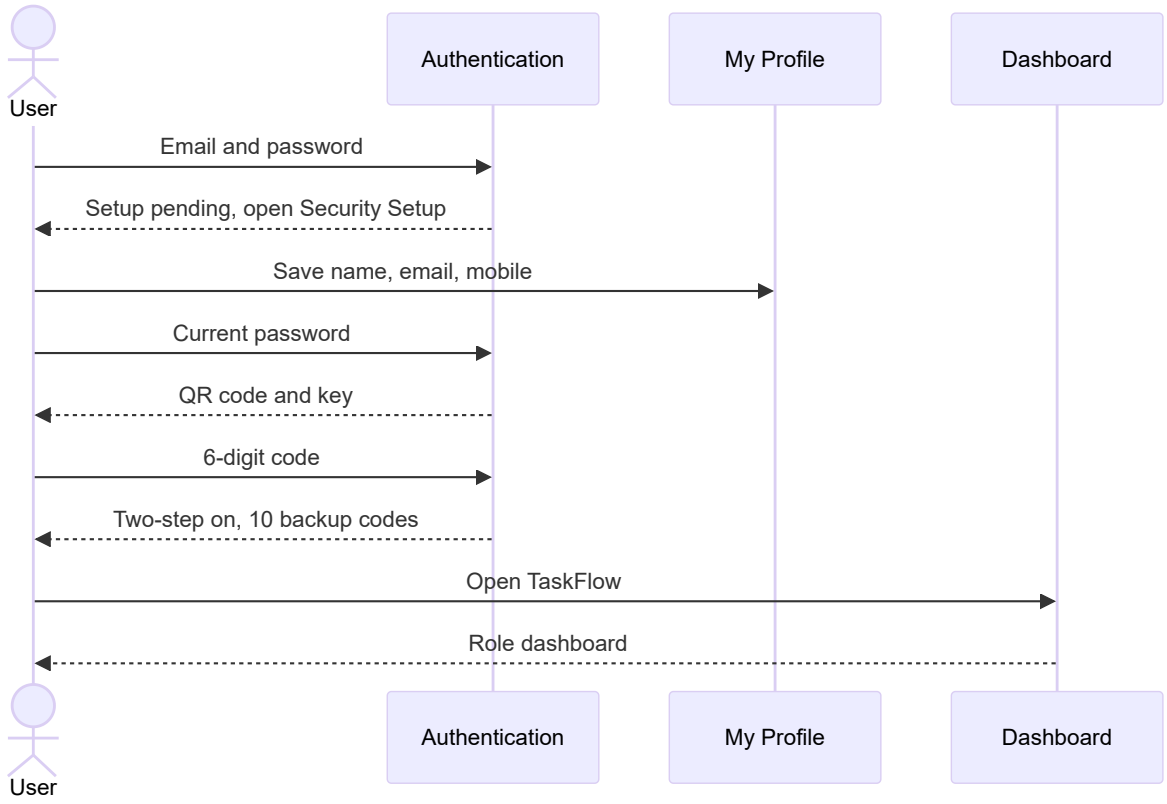
Step	Module	Actor / System	Action	FR reference
1	NOTIF	System	Reminder engine runs at start-up and every 60 seconds	FR-NOTIF ← REQ-NOTIF-28
2	REM	System	Reads "Send deadline reminders", deadline-day reminder, Mail delivery	FR-NOTIF ← REQ-NOTIF-29; FR-REM ← REQ-REM-03, REQ-REM-17
3	TASK	System	Takes open tasks due within 32 days or overdue	FR-NOTIF ← REQ-NOTIF-30

Step	Module	Actor / System	Action	FR reference
4	APP	System	Decides today's reminder type in the application timezone (UTC if unrecognised)	FR-NOTIF ← REQ-NOTIF-31 – REQ-NOTIF-36; FR-APP ← REQ-APP-07, REQ-APP-08
5	NOTIF	System	Resolves recipients plus creator, active only; skips already delivered	FR-NOTIF ← REQ-NOTIF-37 – REQ-NOTIF-39
6	NOTIF	System	Writes the in-app reminder (N-06 – N-09) and the reminder history	FR-NOTIF ← REQ-NOTIF-41, REQ-NOTIF-42, REQ-NOTIF-44; FR-REM ← REQ-REM-09
7	MAIL	System	When Mail delivery is on, places a mail from "TaskFlow" in each Inbox (M-01)	FR-NOTIF ← REQ-NOTIF-43; FR-REM ← REQ-REM-08
8	NOTIF	System	Writes nothing for a task completed, cancelled, deleted or rescheduled during the run; a failed run is logged	FR-NOTIF ← REQ-NOTIF-40, REQ-NOTIF-45



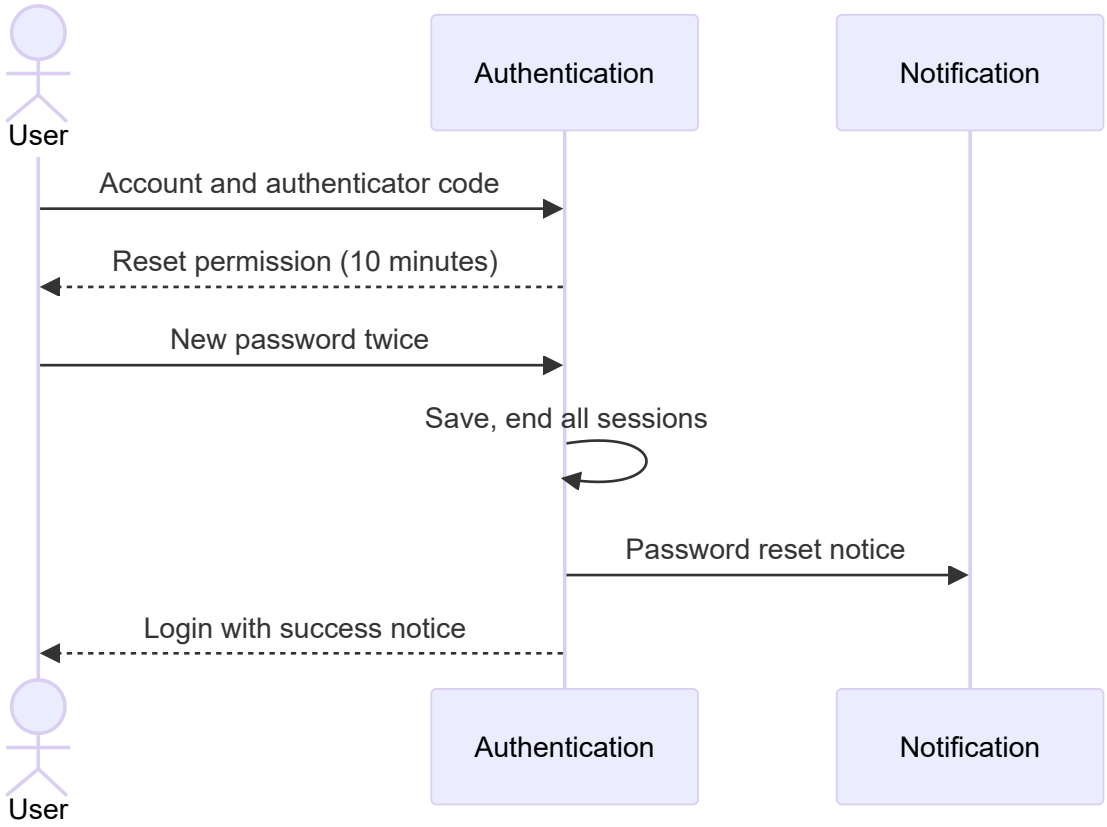
5.3 Sign-in with Two-Step Verification and First-Time Setup

Step	Module	Actor / System	Action	FR reference
1	EMP	Administrator	Creates the account (Employee role by default)	FR-EMP ← REQ-EMP-30, REQ-EMP-36
2	AUTH	User	Signs in with email and password	FR-AUTH-01 – FR-AUTH-04
3	AUTH	System	Setup pending → opens Security Setup; every other screen and function closed	FR-AUTH-10, FR-AUTH-29
4	AUTH, PROF	User	Saves details (own profile functions), checks password, scans QR, verifies code, saves backup codes	FR-AUTH-31 – FR-AUTH-38; FR-PROF ← REQ-PROF-12
5	DASH	System	Opens the role dashboard	FR-AUTH-37, FR-AUTH-09; FR-DASH ← REQ-DASH-02
6	AUTH	User	Later sign-ins: password, then 6-digit code	FR-AUTH-08, FR-AUTH-12 – FR-AUTH-14



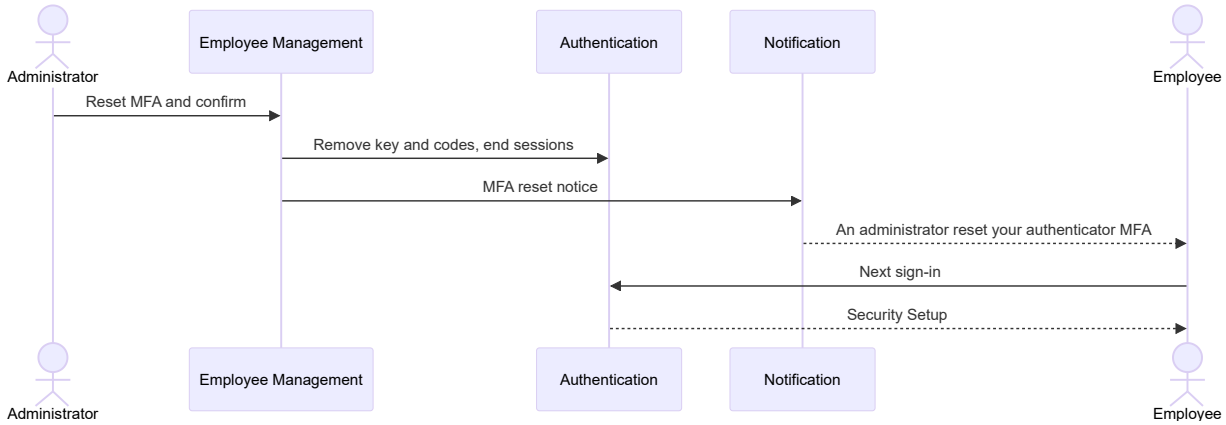
5.4 Self-Service Account Recovery

Step	Module	Actor / System	Action	FR reference
1	AUTH	User	Enters account and authenticator code	FR-AUTH-16 – FR-AUTH-21
2	AUTH	System	Grants a 10-minute single-use reset permission	FR-AUTH-22
3	AUTH	User	Sets a new password	FR-AUTH-23
4	AUTH	System	Ends every session of the user	FR-AUTH-24, FR-AUTH-48
5	NOTIF	System	Sends the System notification (N-12)	FR-AUTH-25; FR-NOTIF ← REQ-NOTIF-25
6	AUTH	User	Signs in with the new password	FR-AUTH-26, FR-AUTH-01



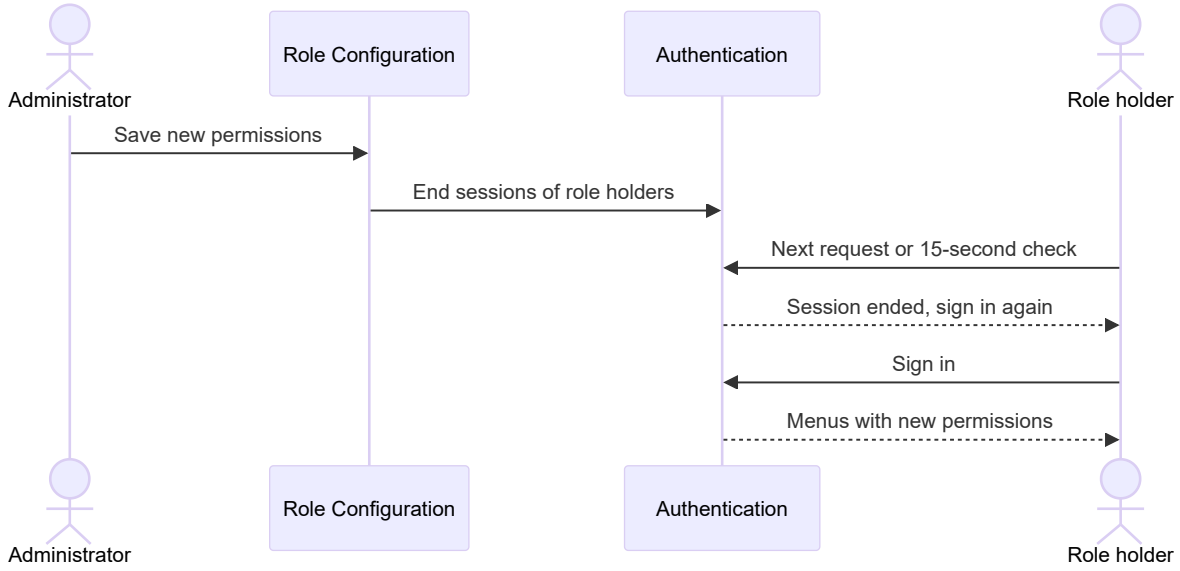
5.5 Lost Authenticator – Administrator MFA Reset

Step	Module	Actor / System	Action	FR reference
1	EMP	Administrator	Opens the employee details panel; **Reset MFA** is shown only with `employees.mfa.reset` and when MFA is on	FR-EMP ← REQ-EMP-27
2	EMP	Administrator	Confirms "Reset two-step verification?"	FR-EMP ← REQ-EMP-28
3	AUTH	System	Removes key and backup codes, changes the security stamp, ends all sessions	FR-AUTH-48
4	NOTIF	System	Sends the System notification (N-11)	FR-EMP ← REQ-EMP-29; FR-NOTIF ← REQ-NOTIF-24
5	AUTH	Employee	Next sign-in → Security Setup	FR-AUTH-10, FR-AUTH-29



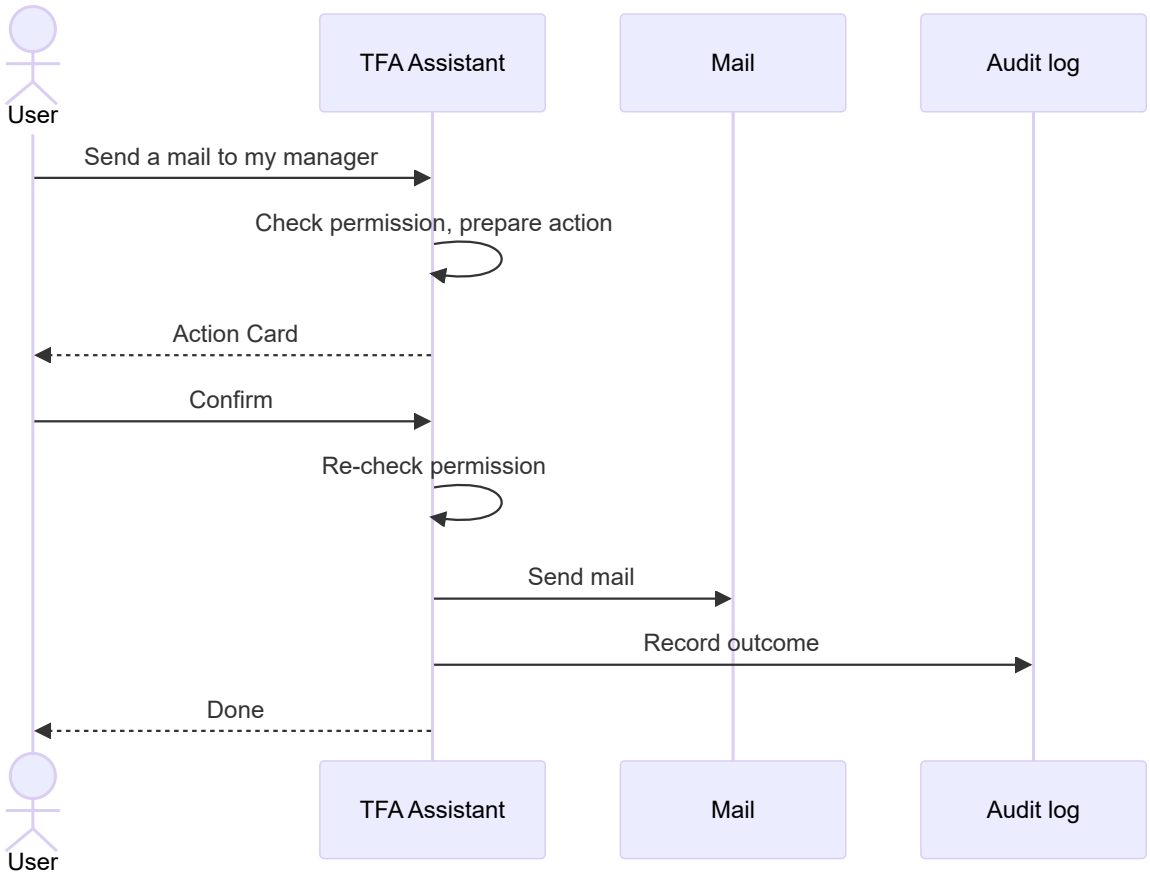
5.6 Role or Permission Change Reaches Signed-in Users

Step	Module	Actor / System	Action	FR reference
1	ROLE	Administrator	Changes a custom role's permissions (system roles are read-only)	FR-ROLE ← REQ-ROLE-06, REQ-ROLE-15
2	ROLE	System	Signs out every holder of that role when permissions changed (not for a rename)	FR-ROLE ← REQ-ROLE-17
3	EMP	Administrator	Or gives an employee another role (Administrator role only with `roles.administrator.assign`)	FR-EMP ← REQ-EMP-37, REQ-EMP-44
4	AUTH	System	Next request and the 15-second check use the current role and permissions; menus update	FR-AUTH-44, FR-AUTH-45, FR-AUTH-50



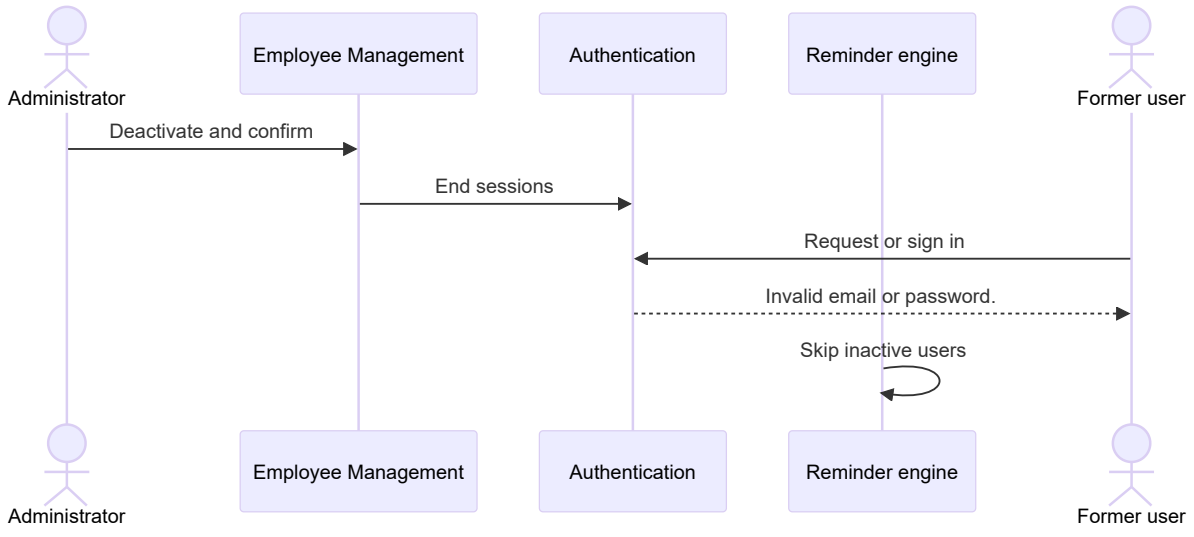
5.7 TFA Prepared Action (confirm before acting)

Step	Module	Actor / System	Action	FR reference
1	TFA	User	Asks TFA, for example to send a mail to the manager	FR-TFA ← REQ-TFA-01, REQ-TFA-27
2	TFA	System	Checks the user's permission and prepares an Action Card (expires in 10 minutes)	FR-TFA ← REQ-TFA-22 – REQ-TFA-24
3	TFA	User	Clicks **Confirm**	FR-TFA ← REQ-TFA-22
4	TFA, MAIL / PROF / EMP	System	Re-checks the permission with the current sign-in and runs the action once (send mail, update own profile, change a user's role)	FR-TFA ← REQ-TFA-23, REQ-TFA-25; FR-PROF ← REQ-PROF-13
5	TFA	System	Records the outcome in the audit log, without message bodies or secrets	FR-TFA ← REQ-TFA-43 – REQ-TFA-45



5.8 Employee Deactivation or Deletion

Step	Module	Actor / System	Action	FR reference
1	EMP	Administrator	Deactivates (with confirmation) or deletes an employee; last active Administrator protected	FR-EMP ← REQ-EMP-13, REQ-EMP-15, REQ-EMP-16, REQ-EMP-18
2	AUTH	System	Ends the person's sessions; sign-in refused with the generic message	FR-EMP ← REQ-EMP-14, REQ-EMP-20; FR-AUTH-03, FR-AUTH-45
3	NOTIF	System	No reminders or notifications to inactive or deleted users	FR-NOTIF ← REQ-NOTIF-27, REQ-NOTIF-37
4	TASK, MAIL	System	Tasks, comments and mail are kept; names still shown	FR-EMP ← REQ-EMP-16; FR-MAIL ← REQ-MAIL-21



6. Reports & Exports

System-level overview; filters, columns and messages are specified in the sub-module FRDs.

Report	Module	Filters	Columns / content	Export	Access
Task Reports – totals and completion summary	RPT	Deadline range, employee, project, status, priority	Total, completed, pending, overdue, completion summary; filtered task rows	CSV `taskflow-report.csv`, at most 50,000 rows, opened without running formulas	`reports.view`; export `reports.export`
Dashboard analytics	DASH	Period chips, date basis (Created / Deadline / Completed), dashboard filters	By status, priority, deadline, time, department, employee, manager, project; drill-down to the counted tasks	–	`dashboard.analytics.view`, `dashboard.filters.use`, scope by `dashboard.view.*`
Employee Workload	WORK	Project, employee	Employee ("High load" badge), projects and %, allocated %, active, high priority, overdue, completed	–	`workload.view`, `workload.filters.use`, `workload.tasks.open`
Activity History	TASK	Activity filters	Event, user, old and new value, time	–	`tasks.activity.view` and task access

Authentication and Legal produce no reports.

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7. Notifications & Messages Catalogue

7.1 In-app notifications (header bell, Notification Center)

Event notifications (N-01 – N-05, N-10 – N-12) are not created while in-app notifications are switched off in Application Settings (REQ-APP-12, REQ-NOTIF-26); reminder notifications (N-06 – N-09) are always created when reminders run (REQ-REM-09). Deleted users are never notified.

Code	Trigger	Recipients	Type shown	Text
N-01	Task created	Assignee	Task assigned	"A task was assigned to you: *title*"
N-02	Task reassigned	New assignee	Task reassigned	"A task was reassigned to you: *title*. Reason: *reason*"
N-03	Task reassigned	Previous assignee, recipients, acting person	Task reassigned	"*Name* reassigned task "*title*" (*code*) from *old* to *new*. Reason: *reason*"
N-04	Comment added, edited or deleted	Assignee, recipients, acting person (active, not deleted)	Comment	"*Name added / edited / deleted* a comment on "*title*": *detail*"

Code	Trigger	Recipients	Type shown	Text
N-05	Worked hours added, updated or cleared	Assignee and recipients (incl. the acting person when a recipient)	System	"*Name action* on task *title*. Total worked now: *n*h of *m*h estimated (*r*h remaining / over by *r*h)."
N-06	First reminder day	Recipients (assignee always) and creator, active only	Deadline reminder	"Reminder: "*title*" is due in *N* days (*dd MMM yyyy, HH:mm*)." ("1 day" when N is 1)
N-07	Second reminder day	Same as N-06	Deadline reminder	Same text as N-06
N-08	Deadline day, before the deadline time	Same as N-06	Deadline reminder	"Reminder: "*title*" is due today at *HH:mm*."
N-09	Each day (Daily) or every 7 days (Weekly) after the deadline while open	Same as N-06	Overdue	"Overdue: "*title*" was due on *dd MMM yyyy, HH:mm*."
N-10	Task status changed	–	–	No notification; recorded in the activity history only
N-11	Administrator MFA reset	Account owner	System	"An administrator reset your authenticator MFA. Sign in with your password and set it up again from Profile."
N-12	Password reset with authenticator	Account owner	System	"Your password was reset with your authenticator app. If this was not you, contact an administrator now."

7.2 Internal Mail generated by TaskFlow

Code	Trigger	Recipients	Sender	Subject	Condition
M-01	Reminder N-06 – N-09	Same as the reminder	"TaskFlow" (system)	"Task reminder: *title*" / "Task due today: *title*" / "Overdue task: *title*"; body ends "This is an automatic reminder from TaskFlow."	"Also deliver reminders to Mail" on
M-02	Comment added, edited or deleted	Same as N-04	"TaskFlow" (system)	"Comment *added / edited / deleted*: *title*"; body ends "This is an automatic message from TaskFlow."	Same Mail delivery setting
M-03	Task reassigned	Active participants of N-02 / N-03	The person who reassigned	"Task reassigned: *title*"	Always (best effort)
M-04	Worked hours changed	Active assignee and recipients	The person who changed the hours	"Worked hours changed: *title*"	Always (best effort)

System mails from "TaskFlow" can only be forwarded, not replied to. A failed notification or mail never undoes the saved task change (it is logged).

7.3 System-level messages

Code	Situation	Text	Where
G-01	Session ended or renewal failed	"Your session has ended, for example because your username or password was changed. Please sign in again."	Login notice
G-02	Credentials changed (own password change, username change)	"Your username or password was changed. Sign in again with your new details."	Login notice
G-03	Password reset	"Your password has been reset. Sign in with your new password."	Login notice
G-04	Setup pending	"Set up two-step verification to continue."	Server answer to any function except sign-in, own profile, MFA setup
G-05	Concurrent change of the same record	"Someone else changed this record while you were saving. Reload it and try again."	Any save
G-06	Unexpected server fault	"Something went wrong. Please try again."	Any function
G-07	Missing permission	"403 Forbidden"; screen shows its fallback error; screen address → own dashboard	Any protected function or screen
G-08	Record of another user (notification, mail, conversation)	Reported as not found	NOTIF, MAIL, TFA
G-09	Too many attempts (recovery, password re-entry)	"Too many attempts. Try again in a few minutes."	Account Recovery, Security Setup, My Profile
G-10	Code lockout	"Too many wrong codes. Try again in a few minutes."	Login, Security Setup, My Profile

8. Data Requirements (functional view)

Data item	Description	Created by	Rules
User account	Employee code, full name, username, email, mobile number, role, active flag, last login	Administrator (EMP); user updates own details (PROF, Security Setup)	Username unique; email unique on `@taskflow.com`, stored in lower case; soft delete
Password	Secret for sign-in	User; Administrator at creation	Stored only as salted hash; 6 – 128 characters; old hashes upgraded at sign-in
Authenticator key	Shared secret for 6-digit codes	System at Security Setup	Stored encrypted; never shown after setup; removed by MFA reset
Backup codes	10 single-use codes `XXXXX-XXXXX`	System when two-step verification turns on	Stored only as hashes; each usable once; shown once
Code lockout state	Wrong-code counter and lock time, last used code step	System	5 wrong codes → 5-minute lock; each code step accepted once

Data item	Description	Created by	Rules
Sign-in step	Proof of correct password for the code step	System	Valid 5 minutes; void after a security stamp change
Renewal token	Session renewal	System at sign-in and renewal	Stored only as hash; 7 days; replaced at each renewal; revoked at sign-out and security changes
Reset permission	Password reset after recovery	System after a valid recovery code	Stored only as hash; 10 minutes; single use; others cancelled when one is used
Security stamp	Session validity marker	System	Changes on password change, password reset, MFA reset
Attempt counters	Recovery and password re-entry throttles	System	Kept in memory; reset at application restart
Owner record	Owner name, year, contact email, rights statement, application name	Fixed in the application	Same values for footer, legal pages and browser page details
Legal documents	Five documents with sections	Fixed in the application	Read-only; "Last updated" 21 October 2026
Other data sets	Tasks, recipients, comments, history, notifications, reminder history, mail, settings, roles, AI settings, assistant data	Sub-modules	See overall BRD section 8 and the sub-module FRDs

9. Non-Functional Requirements (functional impact)

ID	Requirement	Affects
NFR-01	Long lists are paged: tasks and notifications 20 rows by default (at most 100), Mail 25 per page.	Task List, Notification Center, Mail
NFR-02	Dashboard and report totals are calculated by the server. ⚠️ To be confirmed: response-time target.	Dashboards, Reports
NFR-03	A CSV export contains at most 50,000 rows.	Reports
NFR-04	Passwords only as salted hashes; new passwords 6 – 128 characters; never shown or logged.	Login, Account Recovery, Security Setup, My Profile, Employee Management (FR-AUTH-04, FR-AUTH-23)
NFR-05	Two-step verification (6-digit time-based codes, 10 backup codes) is required and enforced by the server.	All signed-in functions (FR-AUTH-28, FR-AUTH-29)
NFR-06	Access pass 30 minutes; renewal cookie 7 days, HttpOnly, Secure, SameSite Strict, replaced at each renewal.	Sessions (FR-AUTH-41 – FR-AUTH-43)
NFR-07	Throttling: 5 wrong codes lock 5 minutes; recovery 5 per account and 20 per address in 5 minutes; password re-entry 5 per 5 minutes; TFA 20 messages per minute per user.	Login, Account Recovery, Security Setup, My Profile, TFA (FR-AUTH-14, FR-AUTH-20, FR-AUTH-32)

ID	Requirement	Affects
NFR-08	Every write is validated by the server and returns field messages; queries are safe against injection.	All forms
NFR-09	Another user's task, mail, notification or conversation is refused or reported as not found.	TASK, MAIL, NOTIF, TFA
NFR-10	AI provider keys and authenticator keys are encrypted with a key ring outside the database; only the last 4 characters of an AI key are shown.	TFA Settings, Security Setup (FR-AUTH-38)
NFR-11	Data stays on the server unless a cloud AI model is chosen; notice on first cloud use; local model is the default.	TFA, TFA Settings, Privacy Policy (FR-LEGAL-10)
NFR-12	Every refused action shows a message; destructive actions and every TFA action ask for confirmation; menus show only permitted entries.	All screens (FR-AUTH-50)
NFR-13	Tasks, history, notifications and reminder history survive a restart.	TASK, NOTIF
NFR-14	Reminder engine: first run at start-up, then every 60 seconds; failed run logged; no duplicates.	Reminder engine
NFR-15	Modular, layered application; structure rules run with the automated tests.	All modules
NFR-16	Modern desktop browsers with responsive layout; voice input needs Chrome or Edge; server on Windows, Linux, macOS.	All screens, TFA voice
NFR-17	All data in one data folder with a documented backup and restore; key ring backed up separately.	Operations, Privacy Policy
NFR-18	Keyboard use and readability. ⚠️ To be confirmed: accessibility target.	All screens
NFR-19	Every request logged without its content (method, path, status, time, user, trace id); refused authorisations and TFA outcomes recorded.	All modules
NFR-20	Works without the internet except for cloud AI providers.	Authentication (offline authenticator), NOTIF, MAIL, TFA

10. Assumptions & Open Issues

Type	Description	Owner
Assumption	TaskFlow serves one organisation on a single application server; attempt counters and the TFA rate limit are in memory and reset on restart.	Jojo George

Type	Description	Owner
Assumption	Every user has a phone with an authenticator app (Google Authenticator, Microsoft Authenticator, Authy or similar).	Jojo George
Assumption	All deadline days are decided in one application timezone (default Asia/Kolkata).	Jojo George
Assumption	Sub-module FRDs number their requirements independently; this document refers to them through their BRD trace.	Jojo George
Open issue	⚠️ To be confirmed: the Login form requires a valid email format although the server also accepts a username (FR-AUTH-01, FR-AUTH-02).	Jojo George
Open issue	⚠️ To be confirmed: the server accepts an unused backup code at the sign-in code step, but the code box takes only 6 characters (4.1.2).	Jojo George
Open issue	⚠️ To be confirmed: password sign-in itself has no attempt limit (only the code step is limited); whether a limit is required (R-06).	Jojo George
Open issue	⚠️ To be confirmed: regenerating backup codes exists as a server function but no screen offers it (REQ-PROF-29).	Jojo George
Open issue	⚠️ To be confirmed: the Reset MFA confirmation says the employee "can sign in with just their password" and points to Profile, while with two-step verification required the employee is sent to Security Setup on the next sign-in.	Jojo George
Open issue	⚠️ To be confirmed: while setup is pending the own-profile functions stay open on the server, but the `/profile` screen itself redirects to Security Setup (BR-03 wording "own profile").	Jojo George
Open issue	⚠️ To be confirmed: an unknown `/legal/...` address goes to Login, so "Legal document not found" may not be reached (FR-LEGAL-07).	Jojo George
Open issue	⚠️ To be confirmed: response-time target (NFR-02) and accessibility target (NFR-18).	Jojo George
Open issue	⚠️ To be confirmed: whether Managers' reports and workload should be limited to their team (R-09).	Jojo George

11. Requirements Traceability Matrix

11.1 Authentication and Legal requirements

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-AUTH-01	FR-AUTH-01, FR-AUTH-02	Login	UC-AUTH-01	AuthServiceLoginTests, AuthEndpointsTests
REQ-AUTH-02	FR-AUTH-03	Login	UC-AUTH-01 (E2)	AuthServiceLoginTests, AuthEndpointsTests

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-AUTH-03	FR-AUTH-04	Login	UC-AUTH-01	AuthServiceLoginTests
REQ-AUTH-04	FR-AUTH-08, FR-AUTH-15	Login, Two-step step	UC-AUTH-01	AuthServiceMfaTests, MfaSignInEndpointsTests
REQ-AUTH-05	FR-AUTH-12	Two-step step	UC-AUTH-01 (E3)	AuthServiceMfaTests, MfaSignInEndpointsTests
REQ-AUTH-06	FR-AUTH-13, FR-AUTH-15	Two-step step	UC-AUTH-01	TotpTests, MfaServiceTests
REQ-AUTH-07	FR-AUTH-14	Two-step step	UC-AUTH-01 (E4)	MfaServiceTests, MfaSignInEndpointsTests
REQ-AUTH-08	FR-AUTH-09, FR-AUTH-51	Login	UC-AUTH-01	AuthEndpointsTests
REQ-AUTH-09	FR-AUTH-10	Login	UC-AUTH-01 (A2)	MfaSetupGateEndpointsTests
REQ-AUTH-10	FR-AUTH-07	Login	UC-AUTH-01	AuthServiceLoginTests
REQ-AUTH-11	FR-AUTH-05	Login	UC-AUTH-01	⚠️ No automated frontend test found
REQ-AUTH-12	FR-AUTH-06	Login	UC-AUTH-01	⚠️ No automated frontend test found
REQ-AUTH-13	FR-AUTH-41, FR-AUTH-42	Sessions	UC-AUTH-04	AuthServiceRefreshAndLogoutTests
REQ-AUTH-14	FR-AUTH-43	Sessions	UC-AUTH-04	AuthServiceRefreshAndLogoutTests
REQ-AUTH-15	FR-AUTH-44, FR-AUTH-50	Sessions	UC-AUTH-04	AuthenticationWiringTests
REQ-AUTH-16	FR-AUTH-45, FR-AUTH-48, FR-AUTH-49	Sessions	UC-AUTH-04	AuthenticationWiringTests, PermissionAuthorizationTests
REQ-AUTH-17	FR-AUTH-46	Sessions	UC-AUTH-05	AuthServiceRefreshAndLogoutTests
REQ-AUTH-18	FR-AUTH-11, FR-AUTH-47	Login, Sessions	UC-AUTH-04 (E1)	⚠️ No automated frontend test found
REQ-AUTH-19	FR-AUTH-16	Account Recovery	UC-AUTH-02	PasswordRecoveryServiceTests, PasswordRecoveryEndpointsTests
REQ-AUTH-20	FR-AUTH-17	Account Recovery	UC-AUTH-02	PasswordRecoveryServiceTests
REQ-AUTH-21	FR-AUTH-18	Account Recovery	UC-AUTH-02	PasswordRecoveryServiceTests
REQ-AUTH-22	FR-AUTH-19	Account Recovery	UC-AUTH-02 (E2)	PasswordRecoveryEndpointsTests
REQ-AUTH-23	FR-AUTH-20	Account Recovery	UC-AUTH-02 (E1)	AttemptLimiterTests, PasswordRecoveryEndpointsTests
REQ-AUTH-24	FR-AUTH-21	Account Recovery	UC-AUTH-02	PasswordRecoveryServiceTests

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-AUTH-25	FR-AUTH-22	Account Recovery	UC-AUTH-02 (E3)	PasswordRecoveryServiceTests
REQ-AUTH-26	FR-AUTH-23	Account Recovery	UC-AUTH-02	PasswordRecoveryEndpointsTests
REQ-AUTH-27	FR-AUTH-24, FR-AUTH-48	Account Recovery, Sessions	UC-AUTH-02	PasswordRecoveryServiceTests
REQ-AUTH-28	FR-AUTH-25	Account Recovery	UC-AUTH-02	PasswordRecoveryServiceTests
REQ-AUTH-29	FR-AUTH-11, FR-AUTH-26	Account Recovery, Login	UC-AUTH-02	⚠️ No automated frontend test found
REQ-AUTH-30	FR-AUTH-27	Account Recovery	UC-AUTH-02 (E3)	⚠️ No automated frontend test found
REQ-AUTH-31	FR-AUTH-28	Security Setup	UC-AUTH-03	MfaSetupGateEndpointsTests
REQ-AUTH-32	FR-AUTH-29, FR-AUTH-30	Security Setup	UC-AUTH-03	MfaSetupGateEndpointsTests
REQ-AUTH-33	FR-AUTH-31, FR-AUTH-40	Security Setup	UC-AUTH-03	MfaSetupGateEndpointsTests
REQ-AUTH-34	FR-AUTH-32	Security Setup	UC-AUTH-03 (E2)	MfaEnrolmentEndpointsTests, MfaStepUpEndpointsTests
REQ-AUTH-35	FR-AUTH-33	Security Setup	UC-AUTH-03	⚠️ No automated frontend test found
REQ-AUTH-36	FR-AUTH-34	Security Setup	UC-AUTH-03 (E3)	MfaServiceTests, MfaEnrolmentEndpointsTests
REQ-AUTH-37	FR-AUTH-35	Security Setup	UC-AUTH-03 (E4)	MfaEnrolmentEndpointsTests
REQ-AUTH-38	FR-AUTH-36	Security Setup	UC-AUTH-03	MfaServiceTests, MfaRecoveryEndpointsTests
REQ-AUTH-39	FR-AUTH-37	Security Setup	UC-AUTH-03	⚠️ No automated frontend test found
REQ-AUTH-40	FR-AUTH-38	Security Setup	UC-AUTH-03	MfaServiceTests
REQ-AUTH-41	FR-AUTH-39	Security Setup	UC-AUTH-03 (A1)	⚠️ No automated frontend test found
REQ-LEGAL-01	FR-LEGAL-01	Legal Document Page	UC-LEGAL-01	⚠️ No automated test found
REQ-LEGAL-02	FR-LEGAL-02	Legal Document Page	UC-LEGAL-01	⚠️ No automated test found
REQ-LEGAL-03	FR-LEGAL-03, FR-LEGAL-04	Legal Document Page	UC-LEGAL-01	⚠️ No automated test found
REQ-LEGAL-04	FR-LEGAL-05	Legal Document Page	UC-LEGAL-01	⚠️ No automated test found
REQ-LEGAL-05	FR-LEGAL-06	Legal Document Page	UC-LEGAL-01	⚠️ No automated test found

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-LEGAL-06	FR-LEGAL-07	Legal Document Page	UC-LEGAL-01 (E1)	⚠️ No automated test found
REQ-LEGAL-07	FR-LEGAL-08	Legal Document Page	UC-LEGAL-01	⚠️ No automated test found
REQ-LEGAL-08	FR-LEGAL-09	Legal Document Page (Privacy)	UC-LEGAL-01	Content review
REQ-LEGAL-09	FR-LEGAL-10	Legal Document Page (Terms, Privacy)	UC-LEGAL-01	Content review
REQ-LEGAL-10	FR-LEGAL-11	Legal Document Page (Disclaimer)	UC-LEGAL-01	Content review
REQ-LEGAL-11	FR-LEGAL-12	Legal Footer	UC-LEGAL-02	⚠️ No automated test found
REQ-LEGAL-12	FR-LEGAL-13	Legal Footer	UC-LEGAL-02	⚠️ No automated test found
REQ-LEGAL-13	FR-LEGAL-14	Legal Footer	UC-LEGAL-02	⚠️ No automated test found
REQ-LEGAL-14	FR-LEGAL-15	Legal Footer	UC-LEGAL-02	⚠️ No automated test found
REQ-LEGAL-15	FR-LEGAL-16	Legal Footer (browser page details)	UC-LEGAL-02	⚠️ No automated test found

11.2 Global business rules

BRD item	FRD requirement(s) in this document	Sub-module FRD requirements (by BRD trace)	Workflow
BR-01	FR-AUTH-29, FR-AUTH-51, FR-LEGAL-02	–	5.3
BR-02	FR-AUTH-03, FR-AUTH-45	FR-EMP ← REQ-EMP-13, REQ-EMP-14, REQ-EMP-20	5.8
BR-03	FR-AUTH-08, FR-AUTH-10, FR-AUTH-12 – FR-AUTH-14, FR-AUTH-28 – FR-AUTH-38	FR-PROF ← REQ-PROF-12, REQ-PROF-28	5.3
BR-04	FR-AUTH-44, FR-AUTH-45, FR-AUTH-49, FR-AUTH-50	FR-ROLE ← REQ-ROLE-11, REQ-ROLE-17	5.6
BR-05	FR-AUTH-45 (Always On read with the role)	FR-ROLE ← REQ-ROLE-06, REQ-ROLE-13, REQ-ROLE-21	5.6
BR-06	FR-AUTH-09 (dashboard by scope)	FR-TASK ← REQ-TASK-01; FR-DASH ← REQ-DASH-27, REQ-DASH-47; FR-WORK ← REQ-WORK-01	5.1
BR-07	–	FR-TASK ← REQ-TASK-23 (code generated at creation)	5.1
BR-08	–	FR-TASK ← REQ-TASK-12, REQ-TASK-21	5.1
BR-09	–	FR-TASK ← REQ-TASK-19	5.1
BR-10	FR-AUTH-49	FR-TASK ← REQ-TASK-09; FR-BOARD ← REQ-BOARD-07	5.1
BR-11	–	FR-BOARD ← REQ-BOARD-08; FR-TASK ← REQ-TASK-52	5.1
BR-12	–	FR-NOTIF ← REQ-NOTIF-36; FR-DASH ← REQ-DASH-27	5.2

BRD item	FRD requirement(s) in this document	Sub-module FRD requirements (by BRD trace)	Workflow
BR-13	–	FR-NOTIF ← REQ-NOTIF-30, REQ-NOTIF-40	5.2
BR-14	–	FR-REM ← REQ-REM-05 – REQ-REM-07, REQ-REM-14, REQ-REM-17	5.2
BR-15	–	FR-NOTIF ← REQ-NOTIF-38	5.2
BR-16	–	FR-NOTIF ← REQ-NOTIF-37	5.2
BR-17	–	FR-NOTIF ← REQ-NOTIF-31; FR-APP ← REQ-APP-07, REQ-APP-08	5.2
BR-18	–	FR-TASK ← REQ-TASK-41 – REQ-TASK-43	5.1
BR-19	–	FR-TASK ← REQ-TASK-52	5.1
BR-20	FR-AUTH-49 (refusal), 7.3 G-08	FR-NOTIF ← REQ-NOTIF-19; FR-MAIL ← REQ-MAIL-18; FR-TFA ← REQ-TFA-16	5.7
BR-21	–	FR-TASK ← REQ-TASK-37; FR-EMP ← REQ-EMP-16	5.8
BR-22	–	FR-DASH ← REQ-DASH-27; FR-RPT ← REQ-RPT-*	5.1
BR-23	FR-AUTH-16 – FR-AUTH-25 (no e-mail or SMS factor)	FR-MAIL ← REQ-MAIL-23 – REQ-MAIL-25; FR-NOTIF ← REQ-NOTIF-43	5.2, 5.4
BR-24	FR-AUTH-31	FR-EMP ← REQ-EMP-30; FR-PROF ← REQ-PROF-06, REQ-PROF-07	5.3
BR-25	FR-AUTH-24, FR-AUTH-48	FR-PROF ← REQ-PROF-21; FR-EMP ← REQ-EMP-28	5.4, 5.5
BR-26	–	FR-EMP ← REQ-EMP-15	5.8
BR-27	–	FR-TFA ← REQ-TFA-22 – REQ-TFA-25, REQ-TFA-43	5.7
BR-28	FR-AUTH-04, FR-AUTH-38	FR-TFA ← REQ-TFA-44; FR-TFAS ← REQ-TFAS-*	5.3
BR-29	FR-LEGAL-01 – FR-LEGAL-16	–	–

11.3 Global acceptance criteria

BRD item	FRD requirement(s) in this document	Sub-module acceptance (verified in the sub-module FRDs)	Use case
AC-01	FR-AUTH-01, FR-AUTH-05 – FR-AUTH-07, FR-AUTH-09	AC-DASH-01	UC-AUTH-01
AC-02	FR-AUTH-03, FR-AUTH-04	AC-EMP-03	UC-AUTH-01
AC-03	FR-AUTH-08, FR-AUTH-12 – FR-AUTH-15	–	UC-AUTH-01
AC-04	FR-AUTH-11, FR-AUTH-41 – FR-AUTH-48	AC-ROLE-11	UC-AUTH-04, UC-AUTH-05
AC-05	FR-AUTH-16 – FR-AUTH-27	AC-NOTIF-09	UC-AUTH-02
AC-06	FR-AUTH-28 – FR-AUTH-40	AC-PROF-12, AC-PROF-13	UC-AUTH-03
AC-07	FR-AUTH-48 (MFA reset ends sessions)	AC-EMP-01 – AC-EMP-13	–
AC-08	–	AC-TASK-06, AC-TASK-09	–
AC-09	–	AC-TASK-01, AC-TASK-12, AC-TASK-13, AC-TASK-16, AC-BOARD-03	–

BRD item	FRD requirement(s) in this document	Sub-module acceptance (verified in the sub-module FRDs)	Use case
AC-10	–	AC-TASK-03, AC-DASH-05, AC-DASH-12, AC-RPT-02	–
AC-11	–	AC-NOTIF-11, AC-REM-08	–
AC-12	–	AC-NOTIF-12	–
AC-13	–	AC-NOTIF-15, AC-REM-05	–
AC-14	–	AC-NOTIF-14, AC-TASK-12	–
AC-15	–	AC-NOTIF-13	–
AC-16	–	AC-TASK-15	–
AC-17	–	AC-WORK-01 – AC-WORK-04	–
AC-18	–	AC-RPT-05, AC-RPT-11	–
AC-19	FR-AUTH-49 – FR-AUTH-51	AC-EMP-02, AC-RPT-01, AC-REM-11, AC-NOTIF-07, AC-MAIL-06	UC-AUTH-04
AC-20	–	AC-NOTIF-13, AC-NOTIF-16	–
AC-21	FR-AUTH-44 (role changes without restart)	AC-REM-09, AC-APP-05, AC-ROLE-05	–
AC-22	–	AC-MAIL-06, AC-MAIL-13	–
AC-23	–	AC-TFA-09, AC-TFA-10, AC-TFA-17	–
AC-24	FR-LEGAL-01 – FR-LEGAL-08	–	UC-LEGAL-01, UC-LEGAL-02
AC-25	FR-LEGAL-09 – FR-LEGAL-11	–	UC-LEGAL-01
AC-26	FR-LEGAL-12 – FR-LEGAL-16	–	UC-LEGAL-02

Untraced BRD items in scope of this document: **0** (41 REQ-AUTH, 15 REQ-LEGAL, 29 BR, 26 AC).

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12. Approvals

Role	Name	Decision	Date
Product owner	Jojo George	Pending	–
Reviewer	To be assigned	Pending	–

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